

HPCL Engineer

Previous Year Paper
(Mechanical)
23 Dec, 2023



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HINDUSTAN PETROLEUM CORPORATION LIMITED
 REGISTERED OFFICE : 17, JAMSHEDJI TATA ROAD, MUMBAI - 400 020.

Participant ID	
Participant Name	
Test Center Name	
Test Date	23/12/2023
Test Time	2:00 PM - 4:30 PM
Subject	Mechanical Engineer

Section : English Language

Q.1 Choose the most appropriate antonym of the underlined word.

In the coming decade, new robotic technologies could further aggravate the problem.

- Ans ☒ 1. vex
☒ 2. provoke
☒ 3. relieve
☒ 4. irk

Question ID : 630680504774
 Option 1 ID : 6306801972639
 Option 2 ID : 6306801972637
 Option 3 ID : 6306801972640
 Option 4 ID : 6306801972638
 Status : Not Answered
 Chosen Option : -

Q.2 Select the most appropriate idiom for the underlined phrase.

That watch was very expensive when I bought it.

- Ans ☒ 1. In a nutshell
☒ 2. Under the rose
☒ 3. Under the weather
☒ 4. Cost an arm and a leg

Question ID : 630680504773
 Option 1 ID : 6306801972634
 Option 2 ID : 6306801972636
 Option 3 ID : 6306801972633
 Option 4 ID : 6306801972635
 Status : Answered
 Chosen Option : 4

Q.3 Select the most appropriate word to fill in the blank.

We _____ make poor assumptions.

- Ans
- ✓ 1. consistently
 - ✗ 2. consistent
 - ✗ 3. more consist
 - ✗ 4. consist

Question ID : 630680504761
 Option 1 ID : 6306801972592
 Option 2 ID : 6306801972591
 Option 3 ID : 6306801972589
 Option 4 ID : 6306801972590

Status : Answered

Chosen Option : 1

Q.4 Choose the word that can substitute the given group of words.

Opinion or doctrine at variance with the orthodox or accepted doctrine

- Ans
- ✓ 1. Heresy
 - ✗ 2. Bourgeois
 - ✗ 3. Plinth
 - ✗ 4. Tenet

Question ID : 630680504766
 Option 1 ID : 6306801972612
 Option 2 ID : 6306801972611
 Option 3 ID : 6306801972610
 Option 4 ID : 6306801972609

Status : Not Answered

Chosen Option : -

Q.5 The question below consists of a set of labelled sentences. Out of the four options given, select the most logical order of the sentences to form a coherent paragraph.

P. Celts believed that on the night before the new year, the boundary between the worlds of the living and the dead became blurred.

Q. Halloween's origins date back to the ancient Celtic festival of Samhain.

R. The Celts, who lived 2,000 years ago, mostly in the area that is now Ireland, the United Kingdom and northern France, celebrated their new year on November 1.

S. This day marked the end of summer and the harvest and the beginning of the dark, cold winter, a time of year that was often associated with human death.

- Ans
- ✗ 1. PQRS
 - ✗ 2. RSPQ
 - ✗ 3. PQSR
 - ✓ 4. QRSP

Question ID : 630680504764
 Option 1 ID : 6306801972604
 Option 2 ID : 6306801972602
 Option 3 ID : 6306801972603
 Option 4 ID : 6306801972601

Status : Answered

Chosen Option : 2



Q.6 Select the most appropriate idiom for the underlined phrase.

I tried my best to get along with her, but I guess she is a difficult person to deal with.

- Ans
- ☒ 1. A wild goose chase
 - ☒ 2. A hard nut to crack
 - ☒ 3. A piece of cake
 - ☒ 4. All ears

Question ID : 630680504772
 Option 1 ID : 6306801972629
 Option 2 ID : 6306801972632
 Option 3 ID : 6306801972631
 Option 4 ID : 6306801972630
 Status : Answered
 Chosen Option : 2

Q.7 Choose the word that can substitute the given group of words.

The sound made by cats

- Ans
- ☒ 1. Grunt
 - ☒ 2. Cheep
 - ☒ 3. Meow
 - ☒ 4. Bell

Question ID : 630680504765
 Option 1 ID : 6306801972606
 Option 2 ID : 6306801972608
 Option 3 ID : 6306801972605
 Option 4 ID : 6306801972607
 Status : Answered
 Chosen Option : 3

Q.8 The question below consists of a set of labelled sentences. Out of the four options given, select the most logical order of the sentences to form a coherent paragraph.

- P. Standing in a farmer's home in China's Henan Province in the summer of 2018, paleontologists Fenghu Han and Haishui Jiang peered down into a box of rounded lumps of rock.
 Q. The farmer had collected the trove near his home in Neixiang County, which is renowned for its dinosaur eggs.
 R. One stony orb in particular caught the scientists' eyes.
 S. About the size and shape of a billiard ball, the fossil was unlike any dinosaur egg they'd seen before.

- Ans
- ☒ 1. PQRS
 - ☒ 2. SPQR
 - ☒ 3. QPSR
 - ☒ 4. RQPS

Question ID : 630680504763
 Option 1 ID : 6306801972599
 Option 2 ID : 6306801972598
 Option 3 ID : 6306801972597
 Option 4 ID : 6306801972600
 Status : Answered
 Chosen Option : 3



Q.9 Choose the most appropriate synonym of the underlined word.

A rustic area that has a refreshing lack of billboards and shopping malls reminds me of my home town.

- Ans
- ☒ 1. open
 - ☒ 2. countryfied
 - ☒ 3. urbane
 - ☒ 4. insecure

Question ID : 630680504775
 Option 1 ID : 6306801972642
 Option 2 ID : 6306801972643
 Option 3 ID : 6306801972644
 Option 4 ID : 6306801972641
 Status : Not Answered
 Chosen Option : -

Q.10 Select the most appropriate word to fill in the blank.

You are mistaken if you think that you are the _____ creature in this world.

- Ans
- ☒ 1. so beautiful
 - ☒ 2. beautiful
 - ☒ 3. more beautiful
 - ☒ 4. most beautiful

Question ID : 630680504762
 Option 1 ID : 6306801972596
 Option 2 ID : 6306801972593
 Option 3 ID : 6306801972594
 Option 4 ID : 6306801972595
 Status : Answered
 Chosen Option : 4

Comprehension:

Read the passage carefully and select the correct answer for the given blanks out of the four alternatives.

Storms are getting stronger elsewhere too. Hurricane Otis recently (1)_____ the Mexican city of Acapulco, for instance. Otis developed from a regular storm into a (2)_____ hurricane in record time, and was the first time in history that a hurricane in the Eastern Pacific made landfall and sustained itself as the strongest "Category 5" storm. Tropical Cyclone Tej (3)_____ the Arabian Peninsula the week before with 480mm of rainfall in the Al-Ghaydah region in Yemen – eight times the annual average.

SubQuestion No : 11

Q.11 Select the most appropriate option for blank _____ (2)

- Ans
- ☒ 1. liable
 - ☒ 2. hounding
 - ☒ 3. huge
 - ☒ 4. timid

Question ID : 630680509623
 Option 1 ID : 6306801991701
 Option 2 ID : 6306801991702
 Option 3 ID : 6306801991703
 Option 4 ID : 6306801991704
 Status : Not Answered
 Chosen Option : -

Comprehension:

Read the passage carefully and select the correct answer for the given blanks out of the four alternatives.

Storms are getting stronger elsewhere too. Hurricane Otis recently (1)_____ the Mexican city of Acapulco, for instance. Otis developed from a regular storm into a (2)_____ hurricane in record time, and was the first time in history that a hurricane in the Eastern Pacific made landfall and sustained itself as the strongest "Category 5" storm. Tropical Cyclone Tej (3)_____ the Arabian Peninsula the week before with 480mm of rainfall in the Al-Ghaydah region in Yemen – eight times the annual average.

SubQuestion No : 12

Q.12 Select the most appropriate option for blank _____ (3)

- Ans
- ☒ 1. masticated
 - ☒ 2. hit
 - ☒ 3. hired
 - ☒ 4. blasted

Question ID : 630680509624

Option 1 ID : 6306801991705

Option 2 ID : 6306801991708

Option 3 ID : 6306801991707

Option 4 ID : 6306801991706

Status : Not Answered

Chosen Option : –

Comprehension:

Read the passage carefully and select the correct answer for the given blanks out of the four alternatives.

Storms are getting stronger elsewhere too. Hurricane Otis recently (1)_____ the Mexican city of Acapulco, for instance. Otis developed from a regular storm into a (2)_____ hurricane in record time, and was the first time in history that a hurricane in the Eastern Pacific made landfall and sustained itself as the strongest "Category 5" storm. Tropical Cyclone Tej (3)_____ the Arabian Peninsula the week before with 480mm of rainfall in the Al-Ghaydah region in Yemen – eight times the annual average.

SubQuestion No : 13

Q.13 Select the most appropriate option for blank _____ (1)

- Ans
- ☒ 1. devastated
 - ☒ 2. manifested
 - ☒ 3. demonstrated
 - ☒ 4. cultivated

Question ID : 630680509622

Option 1 ID : 6306801991700

Option 2 ID : 6306801991697

Option 3 ID : 6306801991699

Option 4 ID : 6306801991698

Status : Not Answered

Chosen Option : –



Comprehension:

Read the following passage and answer the questions given below.

Because birthday balloons are cheap and colorful, children like to buy and play with them. Balloons were first used for scientific experiments and transportation, but it wasn't long before it found its way into people's hands as an instrument of fun. Michael Faraday made the first rubber balloon in 1824, which was for use in his laboratory. Faraday achieved this by laying two sheets of rubber one on another, then put flour in the middle to prevent them joining all the way through and then sealed them together at the edges. Faraday found that upon filling the balloons with hydrogen, they were able to ascend to a considerable level, however he was disappointed by the fact that the hydrogen kept escaping. The following year, Thomas Hancock, a British inventor and rubber manufacturer, started selling balloon making kits. By 1847 British entrepreneurs were already selling the first balloons made of vulcanized rubber, which was tougher and was more elastic.

SubQuestion No : 14

Q.14 What can be the suitable title for the given passage?

- Ans
- ☒ 1. Birthdays and balloons
 - ☒ 2. Balloons for fun
 - ☒ 3. Craze for balloons and their history
 - ☒ 4. Rubber industry

Question ID : 630680504768
 Option 1 ID : 6306801972613
 Option 2 ID : 6306801972616
 Option 3 ID : 6306801972615
 Option 4 ID : 6306801972614

Status : Not Answered

Chosen Option : -

Comprehension:

Read the following passage and answer the questions given below.

Because birthday balloons are cheap and colorful, children like to buy and play with them. Balloons were first used for scientific experiments and transportation, but it wasn't long before it found its way into people's hands as an instrument of fun. Michael Faraday made the first rubber balloon in 1824, which was for use in his laboratory. Faraday achieved this by laying two sheets of rubber one on another, then put flour in the middle to prevent them joining all the way through and then sealed them together at the edges. Faraday found that upon filling the balloons with hydrogen, they were able to ascend to a considerable level, however he was disappointed by the fact that the hydrogen kept escaping. The following year, Thomas Hancock, a British inventor and rubber manufacturer, started selling balloon making kits. By 1847 British entrepreneurs were already selling the first balloons made of vulcanized rubber, which was tougher and was more elastic.

SubQuestion No : 15

Q.15 Which of the following words means the same as the word 'ascend' used in the passage?

- Ans
- ☒ 1. Rise
 - ☒ 2. Slide
 - ☒ 3. Dip
 - ☒ 4. Dive

Question ID : 630680504771
 Option 1 ID : 6306801972625
 Option 2 ID : 6306801972627
 Option 3 ID : 6306801972626
 Option 4 ID : 6306801972628

Status : Not Answered

Chosen Option : -

Comprehension:

Read the following passage and answer the questions given below.

Because birthday balloons are cheap and colorful, children like to buy and play with them. Balloons were first used for scientific experiments and transportation, but it wasn't long before it found its way into people's hands as an instrument of fun. Michael Faraday made the first rubber balloon in 1824, which was for use in his laboratory. Faraday achieved this by laying two sheets of rubber one on another, then put flour in the middle to prevent them joining all the way through and then sealed them together at the edges. Faraday found that upon filling the balloons with hydrogen, they were able to ascend to a considerable level, however he was disappointed by the fact that the hydrogen kept escaping. The following year, Thomas Hancock, a British inventor and rubber manufacturer, started selling balloon making kits. By 1847 British entrepreneurs were already selling the first balloons made of vulcanized rubber, which was tougher and was more elastic.

SubQuestion No : 16

Q.16 Which of the following statements is NOT correct regarding Thomas Hancock?

- Ans
- ☒ 1. He made kits from balloons.
 - ☒ 2. He was a rubber manufacturer.
 - ☒ 3. He made the first rubber balloon.
 - ☒ 4. He was a British inventor.

Question ID : 630680504770
 Option 1 ID : 6306801972623
 Option 2 ID : 6306801972622
 Option 3 ID : 6306801972624
 Option 4 ID : 6306801972621
 Status : Not Answered
 Chosen Option : -

Comprehension:

Read the following passage and answer the questions given below.

Because birthday balloons are cheap and colorful, children like to buy and play with them. Balloons were first used for scientific experiments and transportation, but it wasn't long before it found its way into people's hands as an instrument of fun. Michael Faraday made the first rubber balloon in 1824, which was for use in his laboratory. Faraday achieved this by laying two sheets of rubber one on another, then put flour in the middle to prevent them joining all the way through and then sealed them together at the edges. Faraday found that upon filling the balloons with hydrogen, they were able to ascend to a considerable level, however he was disappointed by the fact that the hydrogen kept escaping. The following year, Thomas Hancock, a British inventor and rubber manufacturer, started selling balloon making kits. By 1847 British entrepreneurs were already selling the first balloons made of vulcanized rubber, which was tougher and was more elastic.

SubQuestion No : 17

Q.17 Where did Michael Faraday use his first rubber balloon?

- Ans
- ☒ 1. His laboratory
 - ☒ 2. His child's school
 - ☒ 3. His home
 - ☒ 4. His neighbourhood

Question ID : 630680504769
 Option 1 ID : 6306801972618
 Option 2 ID : 6306801972620
 Option 3 ID : 6306801972619
 Option 4 ID : 6306801972617
 Status : Not Answered
 Chosen Option : -

Section : Quantitative Aptitude

Q.1 The ratio of cost of bottle and lunch box are 9 : 5. The cost of bottle is Rs. 400 more than cost of lunch box. What is the cost of lunch box?

- Ans
- ☒ 1. Rs. 900
 - ☒ 2. Rs. 500
 - ☒ 3. Rs. 300
 - ☒ 4. Rs. 700

Question ID : 630680505113
 Option 1 ID : 6306801973981
 Option 2 ID : 6306801973984
 Option 3 ID : 6306801973982
 Option 4 ID : 6306801973983
 Status : Answered
 Chosen Option : 2

Q.2 The product of the three consecutive odd numbers is 6783. What is the sum of the three numbers?

- Ans
- ☒ 1. 57
 - ☒ 2. 43
 - ☒ 3. 49
 - ☒ 4. 55

Question ID : 630680505105
 Option 1 ID : 6306801973950
 Option 2 ID : 6306801973951
 Option 3 ID : 6306801973952
 Option 4 ID : 6306801973949
 Status : Not Answered
 Chosen Option : -

Q.3 What is the value of $\sqrt{\frac{1.44 \times 1.44 + 0.26 \times 0.26 + 0.52 \times 1.44}{2.56 \times 2.56 + 0.06 \times 0.06 - 0.12 \times 2.56}}$?

- Ans
- ☒ 1. 0.78
 - ☒ 2. 0.68
 - ☒ 3. 0.58
 - ☒ 4. 0.98

Question ID : 630680505104
 Option 1 ID : 6306801973946
 Option 2 ID : 6306801973948
 Option 3 ID : 6306801973947
 Option 4 ID : 6306801973945
 Status : Not Answered
 Chosen Option : -

Q.4 Pipe R alone can fill a tank in 144 hours and Pipe S alone can fill the same tank in 216 hours. If both the pipes are opened on alternate hours one at a time and pipe S is opened for first hour, then in how much time the tank will be completely full?

- Ans** ☒ 1. 173 hours
☒ 2. 150 hours
☒ 3. 180 hours
☒ 4. 153 hours

Question ID : 630680505114
 Option 1 ID : 6306801973986
 Option 2 ID : 6306801973985
 Option 3 ID : 6306801973987
 Option 4 ID : 6306801973988
 Status : Not Answered
 Chosen Option : -

Q.5 The average of 24 consecutive even numbers is 133. If the next even number is also considered, then what will be the new average?

- Ans** ☒ 1. 134
☒ 2. 138
☒ 3. 140
☒ 4. 132

Question ID : 630680505097
 Option 1 ID : 6306801973919
 Option 2 ID : 6306801973918
 Option 3 ID : 6306801973917
 Option 4 ID : 6306801973920
 Status : Not Answered
 Chosen Option : -

Q.6 Rs. 8500 is lent at the rate of 27 percent per annum on compound interest (compounded in every four months). What will be the compound interest of 8 months?

- Ans** ☒ 1. Rs. 1598.85
☒ 2. Rs. 1427.85
☒ 3. Rs. 1450.65
☒ 4. Rs. 1648.75

Question ID : 630680505100
 Option 1 ID : 6306801973930
 Option 2 ID : 6306801973931
 Option 3 ID : 6306801973932
 Option 4 ID : 6306801973929
 Status : Not Answered
 Chosen Option : -

Q.7 The perimeter of the top of a rectangular table is 22 metre and its area is 30 metre². What is the length of its diagonal?

- Ans
- ☒ 1. $\sqrt{37}$ metres
 - ☒ 2. $\sqrt{71}$ metres
 - ☒ 3. $\sqrt{61}$ metres
 - ☒ 4. $\sqrt{51}$ metres

Question ID : 630680505125
 Option 1 ID : 6306801974031
 Option 2 ID : 6306801974032
 Option 3 ID : 6306801974029
 Option 4 ID : 6306801974030
 Status : Answered
 Chosen Option : 3

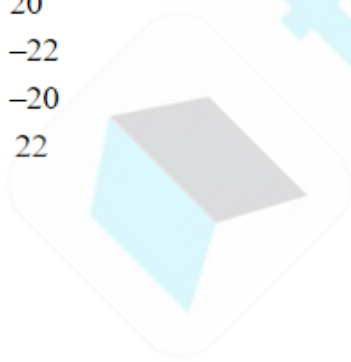
Q.8 The ratio of selling price to the cost price is 6 : 11. What is the loss percentage?

- Ans
- ☒ 1. 42.71 percent
 - ☒ 2. 45.45 percent
 - ☒ 3. 50.68 percent
 - ☒ 4. 52.68 percent

Question ID : 630680505109
 Option 1 ID : 6306801973965
 Option 2 ID : 6306801973967
 Option 3 ID : 6306801973968
 Option 4 ID : 6306801973966
 Status : Answered
 Chosen Option : 2

Q.9 If $f(x) = x^2 + 2x + 7$, then what is the remainder when $f(x)$ is divided by $x + 5$?

- Ans
- ☒ 1. 20
 - ☒ 2. -22
 - ☒ 3. -20
 - ☒ 4. 22



Question ID : 630680505122
 Option 1 ID : 6306801974019
 Option 2 ID : 6306801974018
 Option 3 ID : 6306801974020
 Option 4 ID : 6306801974017
 Status : Answered
 Chosen Option : 4



Q.10 Pankaj sells two calculators at the rate of Rs. 594 per calculator. He earns a profit of 10 percent on one calculator and loses 10 percent on the other calculator. What will be the total profit or loss in whole transaction?

- Ans**
- ☒ 1. Rs. 10 profit
 - ☒ 2. Rs. 12 loss
 - ☒ 3. Rs. 18 profit
 - ☒ 4. Rs. 16 loss

Question ID : 630680505110
 Option 1 ID : 6306801973972
 Option 2 ID : 6306801973971
 Option 3 ID : 6306801973970
 Option 4 ID : 6306801973969
 Status : Not Answered
 Chosen Option : -

Q.11 If $\cos \theta = \frac{3}{5}$, then what is the value of $\frac{1+\tan \theta}{1-\tan \theta}$?

- Ans**
- ☒ 1. -6
 - ☒ 2. 6
 - ☒ 3. -7
 - ☒ 4. 7

Question ID : 630680505129
 Option 1 ID : 6306801974048
 Option 2 ID : 6306801974047
 Option 3 ID : 6306801974046
 Option 4 ID : 6306801974045
 Status : Answered
 Chosen Option : 3

Q.12 Rohit donates 35 percent of his income to a school and deposits 45 percent of the remainder in his bank. If he has Rs. 3575 now, then what is the income of Rohit?

- Ans**
- ☒ 1. Rs. 12000
 - ☒ 2. Rs. 13000
 - ☒ 3. Rs. 10000
 - ☒ 4. Rs. 11000

Question ID : 630680505108
 Option 1 ID : 6306801973963
 Option 2 ID : 6306801973964
 Option 3 ID : 6306801973961
 Option 4 ID : 6306801973962
 Status : Answered
 Chosen Option : 3



Q.13 A bus moves at the speed of 68 km/hr after repairing and moves at the speed of 34 km/hr before repairing. It covers a distance of K km in 8 hours after repairing. How much time will it take to cover a distance of K/4 km before repairing?

- Ans**
- ☒ 1. 5 hours
 - ☒ 2. 6 hours
 - ☒ 3. 8 hours
 - ☒ 4. 4 hours

Question ID : 630680505116
 Option 1 ID : 6306801973996
 Option 2 ID : 6306801973994
 Option 3 ID : 6306801973995
 Option 4 ID : 6306801973993
 Status : Answered
 Chosen Option : 4

Q.14 The diagonals of a rhombus are 8 cm and 10 cm. What is the area of the rhombus?

- Ans**
- ☒ 1. 20 cm²
 - ☒ 2. 40 cm²
 - ☒ 3. 80 cm²
 - ☒ 4. 30 cm²

Question ID : 630680505126
 Option 1 ID : 6306801974036
 Option 2 ID : 6306801974034
 Option 3 ID : 6306801974033
 Option 4 ID : 6306801974035
 Status : Answered
 Chosen Option : 2

Q.15 S1, S2 and S3 alone can complete a work in 12 days, 18 days and 24 days respectively. They received Rs. 3024 to complete the work. They begin the work together but S1 leaves the work after 3 days and S3 leaves the work 7 days before the completion of the work. S2 has completed the remaining work alone. What is the share of S2 from the total money?

- Ans**
- ☒ 1. Rs. 2000
 - ☒ 2. Rs. 1900
 - ☒ 3. Rs. 1800
 - ☒ 4. Rs. 1500

Question ID : 630680505115
 Option 1 ID : 6306801973991
 Option 2 ID : 6306801973989
 Option 3 ID : 6306801973990
 Option 4 ID : 6306801973992
 Status : Not Answered
 Chosen Option : -



Q.16 If $m = 1500$ and $n = 2000$, then m is how much percentage less than n ?

- Ans
- ☒ 1. 12.5 percent
 - ☒ 2. 25 percent
 - ☒ 3. 10 percent
 - ☒ 4. 50 percent

Question ID : 630680505107
 Option 1 ID : 6306801973959
 Option 2 ID : 6306801973957
 Option 3 ID : 6306801973960
 Option 4 ID : 6306801973958
 Status : Answered
 Chosen Option : 2

Q.17 What is the value of $24 \times 16 + 15 \times 16 + 17 \times 16 - 23 \times 21 - 24 \times 15 + 25 \times 26 - 9 \times 8$?

- Ans
- ☒ 1. 531
 - ☒ 2. 631
 - ☒ 3. 641
 - ☒ 4. 541

Question ID : 630680505106
 Option 1 ID : 6306801973956
 Option 2 ID : 6306801973955
 Option 3 ID : 6306801973954
 Option 4 ID : 6306801973953
 Status : Not Answered
 Chosen Option : -

Q.18 A car covers a distance of 35 km in 42 minutes. If its speed is decreased by 30 km/hr, then what will be the time taken by it to cover the same distance?

- Ans
- ☒ 1. 105 minutes
 - ☒ 2. 70 minutes
 - ☒ 3. 85 minutes
 - ☒ 4. 120 minutes

Question ID : 630680505117
 Option 1 ID : 6306801973998
 Option 2 ID : 6306801974000
 Option 3 ID : 6306801973999
 Option 4 ID : 6306801973997
 Status : Answered
 Chosen Option : 1



Q.19 Amir gave two successive discounts of 10 percent and 20 percent on a pencil. If the marked price of the pencil is Rs. 9850, then what is the value of discount?

- Ans ☒ 1. Rs. 2758
☒ 2. Rs. 2688
☒ 3. Rs. 3128
☒ 4. Rs. 2698

Question ID : 630680505111
 Option 1 ID : 6306801973973
 Option 2 ID : 6306801973974
 Option 3 ID : 6306801973975
 Option 4 ID : 6306801973976
 Status : Answered
 Chosen Option : 1

Q.20 18 people sit around a circular table. What is the probability that 2 particular persons will always sit together?

- Ans ☒ 1. $\frac{3}{17}$
☒ 2. $\frac{2}{17}$
☒ 3. $\frac{1}{8}$
☒ 4. $\frac{5}{17}$

Question ID : 630680505121
 Option 1 ID : 6306801974013
 Option 2 ID : 6306801974014
 Option 3 ID : 6306801974015
 Option 4 ID : 6306801974016
 Status : Answered
 Chosen Option : 3

Q.21 If a, b and c are the roots of the cubic equation $7x^3 + 4x - 7 = 0$, then what is the value of $a^3 + b^3 + c^3$?

- Ans ☒ 1. -3
☒ 2. 7
☒ 3. 3
☒ 4. 21

Question ID : 630680505123
 Option 1 ID : 6306801974024
 Option 2 ID : 6306801974023
 Option 3 ID : 6306801974022
 Option 4 ID : 6306801974021
 Status : Not Answered
 Chosen Option : -



Q.22 If $\theta = 30$ degree, then what is the value of $\frac{\tan \theta + 1}{\cot \theta + 1}$?

- Ans
- ☒ 1. $\sqrt{3}$
 - ☒ 2. $\sqrt{3} + 1$
 - ☒ 3. $\sqrt{3} + 2$
 - ☒ 4. $\frac{1}{\sqrt{3}}$

Question ID : 630680505130
 Option 1 ID : 6306801974049
 Option 2 ID : 6306801974051
 Option 3 ID : 6306801974052
 Option 4 ID : 6306801974050
 Status : Answered
 Chosen Option : 4

Q.23 The table given below shows the marked price and value of discount of 7 articles.

Articles	Marked price	Discount
D	1100	500
E	700	200
F	900	500
G	600	400
H	400	300
I	500	200
J	1000	200

Selling price = Marked price – Discount

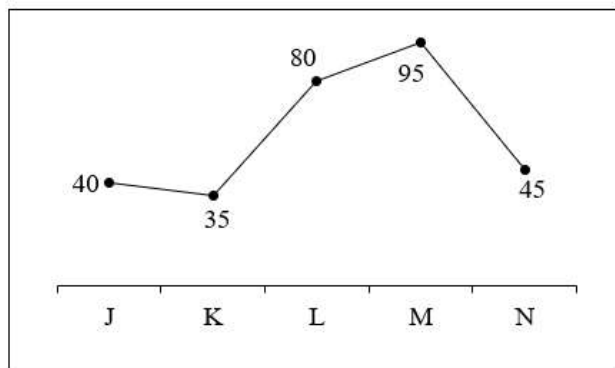
What is the difference between the average selling price and the average marked price of all the articles?

- Ans
- ☒ 1. 328.57
 - ☒ 2. 389.21
 - ☒ 3. 292.57
 - ☒ 4. 240.85

Question ID : 630680505103
 Option 1 ID : 6306801973943
 Option 2 ID : 6306801973941
 Option 3 ID : 6306801973942
 Option 4 ID : 6306801973944
 Status : Answered
 Chosen Option : 1



Q.24 The line chart given below shows the number of monuments in 5 different cities.



What is the average number of monuments in all the cities?

- Ans
- ☒ 1. 31
 - ☒ 2. 63
 - ☒ 3. 47
 - ☒ 4. 59

Question ID : 630680505102
 Option 1 ID : 6306801973940
 Option 2 ID : 6306801973939
 Option 3 ID : 6306801973938
 Option 4 ID : 6306801973937

Status : Answered

Chosen Option : 4

Q.25 In the given set of equation, $x + 2y + 3z = 10$, $2x + 3y + z = 13$, $x + y + 2z = 7$, what is the solution of set?

- Ans
- ☒ 1. $x = 2, y = 3, z = 1$
 - ☒ 2. $x = -1, y = 2, z = 3$
 - ☒ 3. $x = 1, y = 2, z = 3$
 - ☒ 4. $x = 3, y = 2, z = 1$

Question ID : 630680505118
 Option 1 ID : 6306801974001
 Option 2 ID : 6306801974002
 Option 3 ID : 6306801974004
 Option 4 ID : 6306801974003

Status : Answered

Chosen Option : 4



Q.26 If $x^2 + \frac{1}{x^2} = 51$, then what is the value of $\frac{x^2-1}{x}$ ($x > 0$)?

- Ans
- ☒ 1. 7
 - ☐ 2. 6
 - ☐ 3. 8
 - ☐ 4. 4

Question ID : 630680505119
 Option 1 ID : 6306801974005
 Option 2 ID : 6306801974006
 Option 3 ID : 6306801974008
 Option 4 ID : 6306801974007
 Status : Answered
 Chosen Option : 1

Q.27 The breadth of a rectangular hall is $\frac{9}{11}$ of its length. If the area of the floor is 99 metre², then what is the difference between the length and breadth of the hall?

- Ans
- ☐ 1. 3 metres
 - ☐ 2. 4 metres
 - ☐ 3. 1 metre
 - ☒ 4. 2 metres

Question ID : 630680505124
 Option 1 ID : 6306801974027
 Option 2 ID : 6306801974025
 Option 3 ID : 6306801974028
 Option 4 ID : 6306801974026
 Status : Answered
 Chosen Option : 4

Q.28 A boy bought 50 chocolates for Rs. 1200. If the average price of 30 chocolates is Rs. 30, then what is the average price of remaining chocolates?

- Ans
- ☐ 1. Rs. 10
 - ☐ 2. Rs. 20
 - ☒ 3. Rs. 15
 - ☐ 4. Rs. 25

Question ID : 630680505098
 Option 1 ID : 6306801973921
 Option 2 ID : 6306801973923
 Option 3 ID : 6306801973922
 Option 4 ID : 6306801973924
 Status : Answered
 Chosen Option : 3



Q.29 The ratio of two numbers is 5 : 2. If both numbers are increased by 6, the ratio becomes 2 : 1. What is the sum of the initial two numbers?

- Ans**
- ☒ 1. 42
 - ☐ 2. 46
 - ☐ 3. 32
 - ☐ 4. 40

Question ID : 630680505112
 Option 1 ID : 6306801973978
 Option 2 ID : 6306801973979
 Option 3 ID : 6306801973980
 Option 4 ID : 6306801973977
 Status : Answered
 Chosen Option : 1

Q.30 In an equilateral triangle STU, inradius is $10\sqrt{3}$ cm. What is the length of side of this equilateral triangle?

- Ans**
- ☐ 1. $40\sqrt{3}$ cm
 - ☐ 2. $36\sqrt{3}$ cm
 - ☒ 3. 60 cm
 - ☐ 4. 30 cm

Question ID : 630680505127
 Option 1 ID : 6306801974037
 Option 2 ID : 6306801974038
 Option 3 ID : 6306801974040
 Option 4 ID : 6306801974039
 Status : Not Answered
 Chosen Option : —

Q.31 The compound interest (compounding half yearly) received on Rs. 20000 for 1 year is Rs. 6450 What is the rate of interest per annum?

- Ans**
- ☐ 1. 20 percent
 - ☐ 2. 45 percent
 - ☒ 3. 30 percent
 - ☐ 4. 25 percent

Question ID : 630680505101
 Option 1 ID : 6306801973935
 Option 2 ID : 6306801973934
 Option 3 ID : 6306801973936
 Option 4 ID : 6306801973933
 Status : Answered
 Chosen Option : 3



Q.32 If a certain sum of money becomes thrice of itself in 15 years 4 months at simple interest, then what will be the annual rate of interest?

- Ans**
- ☒ 1. 17.28 percent
 - ☒ 2. 15.42 percent
 - ☒ 3. 13.04 percent
 - ☒ 4. 16.84 percent

Question ID : 630680505099
 Option 1 ID : 6306801973926
 Option 2 ID : 6306801973928
 Option 3 ID : 6306801973927
 Option 4 ID : 6306801973925
 Status : Not Answered
 Chosen Option : -

Q.33 Radii of two circles are 10 cm and 2 cm. Distance between their centre is 17 cm. What is the length of the Direct common tangent?

- Ans**
- ☒ 1. 12 cm
 - ☒ 2. 16 cm
 - ☒ 3. 14 cm
 - ☒ 4. 15 cm

Question ID : 630680505128
 Option 1 ID : 6306801974041
 Option 2 ID : 6306801974044
 Option 3 ID : 6306801974043
 Option 4 ID : 6306801974042
 Status : Answered
 Chosen Option : 4

Q.34 If three dices are thrown simultaneously, then what is the probability of getting a sum of 4?

- Ans**
- ☒ 1. $\frac{1}{48}$
 - ☒ 2. $\frac{1}{36}$
 - ☒ 3. $\frac{1}{54}$
 - ☒ 4. $\frac{1}{72}$



Question ID : 630680505120
 Option 1 ID : 6306801974012
 Option 2 ID : 6306801974009
 Option 3 ID : 6306801974011
 Option 4 ID : 6306801974010
 Status : Answered
 Chosen Option : 2

Section : Intellectual Potential Test

Q.1 Consider the five three letter words given below and the answer the questions based on it.

(Left) NOR RST IKL RPN KJA (right)

The given words are arranged in the alphabetical order from left to right. Which word will occupy the third place from right?

- Ans
- ☒ 1. RST
 - ☒ 2. NOR
 - ☒ 3. KJA
 - ☒ 4. RPN

Question ID : 630680504967
 Option 1 ID : 6306801973398
 Option 2 ID : 6306801973397
 Option 3 ID : 6306801973400
 Option 4 ID : 6306801973399

Status : Answered

Chosen Option : 2

Q.2 In the question given below, relationships between different elements are shown in the statements. These statements are followed by two conclusions. Give answer:

Statements:

I. $X = Y = Z$

II. $Y > A \geq B \leq D$

Conclusions:

I. $X > B$

II. $X < D$

- Ans
- ☒ 1. If both conclusions I and II are true
 - ☒ 2. If only conclusion I is true
 - ☒ 3. If only conclusion II is true
 - ☒ 4. If neither conclusion I nor conclusion II is true

Question ID : 630680504972
 Option 1 ID : 6306801973420
 Option 2 ID : 6306801973417
 Option 3 ID : 6306801973418
 Option 4 ID : 6306801973419

Status : Answered

Chosen Option : 1

Q.3 If ' $\times$ ' means ' $-$ ', ' $-$ ' means ' $+$ ', ' $+$ ' means ' $\div$ ' and ' $\div$ ' means ' $\times$ ', then $60 - 30 + 5 \times 7 \div 8 = ?$

- Ans
- ☒ 1. 20
 - ☒ 2. 18
 - ☒ 3. 10
 - ☒ 4. 5

Question ID : 630680504982
 Option 1 ID : 6306801973458
 Option 2 ID : 6306801973460
 Option 3 ID : 6306801973457
 Option 4 ID : 6306801973459

Status : Answered

Chosen Option : 3



Q.4 P, Q, R, S, T, U and V are seven friends sitting in a single row facing North. S is to the immediate right of R. T and P are neighbours of U. Q is to the immediate left of R and on second place from left most end. P is at the right most end. Who is sitting on the immediate right of S?

- Ans**
- ☒ 1. U
 - ☒ 2. P
 - ☒ 3. V
 - ☒ 4. T

Question ID : 630680504955
 Option 1 ID : 6306801973352
 Option 2 ID : 6306801973350
 Option 3 ID : 6306801973351
 Option 4 ID : 6306801973349
 Status : Answered
 Chosen Option : 4

Q.5 Select that missing pair (Fifth number : Sixth number) which is related in the same way as the first number is related to the second number and third number is related to fourth number. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)

17 : 136 :: 21 : 168 :: ?

- Ans**
- ☒ 1. 10 : 45
 - ☒ 2. 15 : 120
 - ☒ 3. 9 : 721
 - ☒ 4. 11 : 80

Question ID : 630680504988
 Option 1 ID : 6306801973482
 Option 2 ID : 6306801973484
 Option 3 ID : 6306801973481
 Option 4 ID : 6306801973483
 Status : Answered
 Chosen Option : 2

Q.6 Ramesh is 7th rank ahead of Ashok in the class of 39 students. If Ashok's Rank is 17th from the last, what is Ramesh's rank from the start?

- Ans**
- ☒ 1. 15
 - ☒ 2. 18
 - ☒ 3. 16
 - ☒ 4. 11

Question ID : 630680504958
 Option 1 ID : 6306801973363
 Option 2 ID : 6306801973364
 Option 3 ID : 6306801973362
 Option 4 ID : 6306801973361
 Status : Answered
 Chosen Option : 3



Q.7 Select that missing pair (Fifth number : Sixth number) which is related in the same way as the first number is related to the second number and third number is related to fourth number. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)

6 : 25 :: 7 : 36 :: ?

- Ans**
- ☒ 1. 10 : 81
 - ☐ 2. 8 : 91
 - ☐ 3. 13 : 121
 - ☐ 4. 11 : 144

Question ID : 630680504987
 Option 1 ID : 6306801973477
 Option 2 ID : 6306801973480
 Option 3 ID : 6306801973479
 Option 4 ID : 6306801973478
 Status : Answered
 Chosen Option : 1

Q.8 'A × B' means 'A is the brother of B', 'A + B' means 'A is the father of B', 'A – B' means 'A is the sister of B' and 'A ÷ B' means 'A is the wife of B'. If $J + K + P \times Z + Q - T$, then how is K related to Z?

- Ans**
- ☐ 1. Husband's mother
 - ☒ 2. Brother's wife
 - ☐ 3. Husband's father
 - ☐ 4. Husband's sister

Question ID : 630680504962
 Option 1 ID : 6306801973379
 Option 2 ID : 6306801973380
 Option 3 ID : 6306801973377
 Option 4 ID : 6306801973378
 Status : Not Answered
 Chosen Option : –

Q.9 If $24 P 8 Q 2 = 28$ and $9 P 6 Q 2 = 12$, then $14 P 7 Q 1 = ?$

- Ans**
- ☐ 1. 19
 - ☐ 2. 20
 - ☒ 3. 21
 - ☐ 4. 11

Question ID : 630680504981
 Option 1 ID : 6306801973454
 Option 2 ID : 6306801973453
 Option 3 ID : 6306801973456
 Option 4 ID : 6306801973455
 Status : Answered
 Chosen Option : 3



Q.10 If '+' means '×', '×' means '÷', '÷' means '+' and '-' means '+', then which of the following equation is NOT correct?

- Ans
- ☒ 1. $24 - 8 + 4 > 30 - 10 \times 4$
 - ☒ 2. $10 \times 4 + 3 > 4 \div 10 \times 3$
 - ☒ 3. $10 + 8 \div 20 = 9 + 11 \div 1$
 - ☒ 4. $40 \times 30 \div 60 = 50 \times 40 \div 60$

Question ID : 630680504983
 Option 1 ID : 6306801973461
 Option 2 ID : 6306801973463
 Option 3 ID : 6306801973462
 Option 4 ID : 6306801973464
 Status : Answered
 Chosen Option : 2

Q.11 In the following question, select the odd letter/letters from the given alternatives.

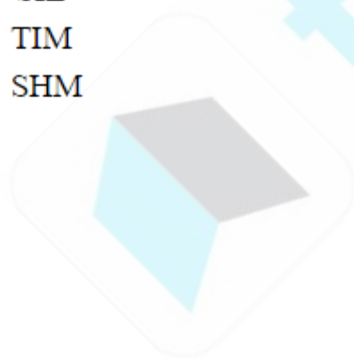
- Ans
- ☒ 1. PONM
 - ☒ 2. FEDC
 - ☒ 3. ZYXW
 - ☒ 4. HGFD

Question ID : 630680504975
 Option 1 ID : 6306801973430
 Option 2 ID : 6306801973432
 Option 3 ID : 6306801973431
 Option 4 ID : 6306801973429
 Status : Answered
 Chosen Option : 4

Q.12 A series is given with one term missing. Select the correct alternative from the given ones that will complete the series.

AQM, YOK, WMI, UKG, ?

- Ans
- ☒ 1. SHE
 - ☒ 2. SIE
 - ☒ 3. TIM
 - ☒ 4. SHM



Question ID : 630680504978
 Option 1 ID : 6306801973444
 Option 2 ID : 6306801973443
 Option 3 ID : 6306801973441
 Option 4 ID : 6306801973442
 Status : Answered
 Chosen Option : 2



Q.13 Rajat walks 8 km towards East. He turns right and walks 4 km. He again turns right and walks 8 km. How far and in which direction is his starting point from the finishing point?

- Ans**
- ☒ 1. 8 km, East
 - ☒ 2. 4 km, South
 - ☒ 3. 4 km, North
 - ☒ 4. 8 km, South

Question ID : 630680504965
 Option 1 ID : 6306801973392
 Option 2 ID : 6306801973389
 Option 3 ID : 6306801973390
 Option 4 ID : 6306801973391
 Status : Answered
 Chosen Option : 3

Q.14 Consider the string given below made up of letters. How many A's are immediately preceded by a B?

AAAABBBAAAABBBBABBBABA

- Ans**
- ☒ 1. 5
 - ☒ 2. 4
 - ☒ 3. 2
 - ☒ 4. 3

Question ID : 630680504969
 Option 1 ID : 6306801973408
 Option 2 ID : 6306801973407
 Option 3 ID : 6306801973405
 Option 4 ID : 6306801973406
 Status : Answered
 Chosen Option : 2

Q.15 A statement is followed by two arguments in the given question. Choose the correct option after deciding which of the arguments is a 'strong' argument and which is a 'weak' argument.

Statement: Should we heed every word our parents say?

Argument:

I. Yes. Parents are never wrong.

II. No. No advice should be followed blindly without mapping its pros and cons.

- Ans**
- ☒ 1. Only argument II is strong
 - ☒ 2. Neither I nor II argument is strong
 - ☒ 3. Both I and II are strong arguments
 - ☒ 4. Only argument I is strong

Question ID : 630680504980
 Option 1 ID : 6306801973450
 Option 2 ID : 6306801973452
 Option 3 ID : 6306801973451
 Option 4 ID : 6306801973449
 Status : Answered
 Chosen Option : 2



Q.16 A statement is followed by a set of conclusions in the given question. Identify which conclusion follows logically from the following statement.

Statement: Students in class A frequently bunk their Mathematics class.

Conclusion:

- I. Students of class A find Mathematics class less intriguing.
- II. Class A students are very intelligent.

Ans

- ☒ 1. Only I follow
- ☐ 2. Only II follows
- ☐ 3. Neither I nor II follows
- ☐ 4. Both I and II follow

Question ID : 630680504979
 Option 1 ID : 6306801973445
 Option 2 ID : 6306801973446
 Option 3 ID : 6306801973448
 Option 4 ID : 6306801973447

Status : Answered

Chosen Option : 1

Q.17 In the following question below are given some statements followed by some conclusions based on those statements. Taking the given statements to be true even if they seem to be at variance from commonly known facts. Read all the conclusions and then decide which of the given conclusion logically follows the given statements.

Statements:

- I. Some paper are ink.
- II. All paper are pencil.

Conclusion:

- I. Some pencil are not ink.
- II. Some pencil are paper.
- III. All ink are pencil.

Ans

- ☐ 1. Only conclusion III follows
- ☐ 2. Both conclusions II and III follows
- ☐ 3. Both conclusions I and III follows
- ☒ 4. Only conclusion II follows

Question ID : 630680504963
 Option 1 ID : 6306801973381
 Option 2 ID : 6306801973384
 Option 3 ID : 6306801973383
 Option 4 ID : 6306801973382

Status : Answered

Chosen Option : 4



Q.18 P, Q, R, S, T, U and V are seven friends sitting in a single row facing North. S is to the immediate right of R. T and P are neighbours of U. Q is to the immediate left of R and on second place from left most end. P is at the right most end. Who is sitting second to the left of T?

- Ans**
- ☒ 1. S
 - ☒ 2. P
 - ☒ 3. Q
 - ☒ 4. R

Question ID : 630680504956
 Option 1 ID : 6306801973353
 Option 2 ID : 6306801973354
 Option 3 ID : 6306801973356
 Option 4 ID : 6306801973355
 Status : Answered
 Chosen Option : 4

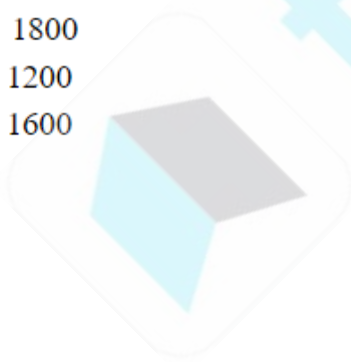
Q.19 In the following question, four letter pairs are given. The letters on left side of (–) is related to the letters on the right side of (–) with some Logic/Rule/Relation. Three are similar on basis of same Logic/Rule/Relation. Select the odd one out from the given alternatives.

- Ans**
- ☒ 1. DRK – CPH
 - ☒ 2. XLE – WJB
 - ☒ 3. NBU – MZR
 - ☒ 4. GLP – HJL

Question ID : 630680504976
 Option 1 ID : 6306801973435
 Option 2 ID : 6306801973436
 Option 3 ID : 6306801973434
 Option 4 ID : 6306801973433
 Status : Answered
 Chosen Option : 4

Q.20 In a certain code language, 'YCD' is written as '900', 'YBA' is written as '150'. What is the code for 'YFD' in that code language?

- Ans**
- ☒ 1. 1500
 - ☒ 2. 1800
 - ☒ 3. 1200
 - ☒ 4. 1600



Question ID : 630680504961
 Option 1 ID : 6306801973374
 Option 2 ID : 6306801973375
 Option 3 ID : 6306801973373
 Option 4 ID : 6306801973376
 Status : Answered
 Chosen Option : 2

Q.21 In the question given below, relationships between different elements are shown in the statements. These statements are followed by two conclusions. Give answer:

Statements:

I. $P < N \leq O \leq Q = R$

II. $R < S < T$

Conclusions:

I. $P < T$

II. $N < S$

- Ans
- ☒ 1. If neither conclusion I nor conclusion II is true
 - ☒ 2. If only conclusion II is true
 - ☒ 3. If both conclusions I and II are true
 - ☒ 4. If only conclusion I is true

Question ID : 630680504971
 Option 1 ID : 6306801973415
 Option 2 ID : 6306801973414
 Option 3 ID : 6306801973416
 Option 4 ID : 6306801973413
 Status : Answered
 Chosen Option : 3

Q.22 In the following question, select the related letters from the given alternatives.

ERT : DQS :: AEM : ?

- Ans
- ☒ 1. ZDL
 - ☒ 2. WAB
 - ☒ 3. XUN
 - ☒ 4. UXU

Question ID : 630680504973
 Option 1 ID : 6306801973424
 Option 2 ID : 6306801973422
 Option 3 ID : 6306801973423
 Option 4 ID : 6306801973421
 Status : Answered
 Chosen Option : 1

Q.23 In a certain code language, 'MODEL' is written as '52', 'JAPAN' is written as '45'. What is the code for 'PUBLIC' in that code language?

- Ans
- ☒ 1. 63
 - ☒ 2. 64
 - ☒ 3. 65
 - ☒ 4. 66

Question ID : 630680504960
 Option 1 ID : 6306801973369
 Option 2 ID : 6306801973372
 Option 3 ID : 6306801973371
 Option 4 ID : 6306801973370
 Status : Answered
 Chosen Option : 4



Q.24 Consider the string given below made up of numbers from 0 to 9. How many odd numbers are immediately followed by another odd number?

1 5 1 6 8 5 9 3 5 9 8 8 4 4 6 5 2 9 4 6

Ans ☒ 1. 6

☐ 2. 7

☐ 3. 5

☐ 4. 4

Question ID : 630680504968

Option 1 ID : 6306801973403

Option 2 ID : 6306801973404

Option 3 ID : 6306801973402

Option 4 ID : 6306801973401

Status : Answered

Chosen Option : 2

Q.25 Each of the following problems has a question and two statements (I) and (II).

A and B are prime numbers. If $A + B$ is odd, then what is the value of A?

I. $A < B$

II. $B - A = 5$

Ans ☒ 1.

If the question can be answered by using either of the statement (I) or (II) alone.

☐ 2.

If the question cannot be answered even by using both the statements together.

☐ 3.

If the question can be answered by using one of the statement alone but cannot be answered by using the other statement alone.

☐ 4.

If the question can be answered by using both the statements together but not by either statement alone.

Question ID : 630680504977

Option 1 ID : 6306801973438

Option 2 ID : 6306801973440

Option 3 ID : 6306801973437

Option 4 ID : 6306801973439

Status : Not Answered

Chosen Option : -

Q.26 Consider the string given below made up of symbols and numbers. How many letters are immediately followed by \$?

#T#T\$JTVSV\$S\$TIV\$J##

Ans ☐ 1. 3

☐ 2. 5

☒ 3. 4

☐ 4. 6

Question ID : 630680504970

Option 1 ID : 6306801973409

Option 2 ID : 6306801973411

Option 3 ID : 6306801973410

Option 4 ID : 6306801973412

Status : Answered

Chosen Option : 3



Q.27 Consider the five three digit numbers given below and the answer the questions based on it.

(Left) 118 231 891 163 555 (right)

What is the sum of the unit digit of the largest and the smallest numbers given?

- Ans
- ☒ 1. 5
 - ☒ 2. 7
 - ☒ 3. 8
 - ☒ 4. 9

Question ID : 630680504966

Option 1 ID : 6306801973396

Option 2 ID : 6306801973393

Option 3 ID : 6306801973394

Option 4 ID : 6306801973395

Status : Answered

Chosen Option : 4

Q.28 In the following question below are given some statements followed by some conclusions based on those statements. Taking the given statements to be true even if they seem to be at variance from commonly known facts. Read all the conclusions and then decide which of the given conclusion logically follows the given statements.

Statements:

- I. Some red are money.
- II. All pink are money.

Conclusion:

- I. No red are pink.
- II. Some money are red.
- III. Some pink are not red.

- Ans
- ☒ 1. Only conclusion II follows
 - ☒ 2. All conclusion follows
 - ☒ 3. Only conclusion I follows
 - ☒ 4. Neither conclusion follows

Question ID : 630680504964

Option 1 ID : 6306801973387

Option 2 ID : 6306801973386

Option 3 ID : 6306801973385

Option 4 ID : 6306801973388

Status : Answered

Chosen Option : 1

Q.29 In the following question, select the missing number from the given series.

180, 177, 168, 141, 60, ?

- Ans
- ☒ 1. -173
 - ☒ 2. -183
 - ☒ 3. -150
 - ☒ 4. 0

Question ID : 630680504984

Option 1 ID : 6306801973466

Option 2 ID : 6306801973465

Option 3 ID : 6306801973467

Option 4 ID : 6306801973468

Status : Answered

Chosen Option : 2



Q.30 Select that missing pair (Third number : Fourth number) which is related in the same way as the first number is related to the second number and fifth number is related to sixth number. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g.13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)

5 : 36 :: ? :: 7 : 64

- Ans**
- ☒ 1. 6 : 25
 - ☒ 2. 15 : 225
 - ☒ 3. 11 : 144
 - ☒ 4. 9 : 81

Question ID : 630680504986
 Option 1 ID : 6306801973474
 Option 2 ID : 6306801973473
 Option 3 ID : 6306801973475
 Option 4 ID : 6306801973476

Status : Answered

Chosen Option : 3

Q.31 In the following question, select the missing number from the given series.

21, 10, 23, 6, 25, ?

- Ans**
- ☒ 1. 7
 - ☒ 2. 8
 - ☒ 3. 2
 - ☒ 4. 5

Question ID : 630680504985
 Option 1 ID : 6306801973471
 Option 2 ID : 6306801973472
 Option 3 ID : 6306801973469
 Option 4 ID : 6306801973470

Status : Answered

Chosen Option : 3

Q.32 Aman ranked 11th from the top and 47th from the bottom in a class. How many students are there in the class?

- Ans**
- ☒ 1. 57
 - ☒ 2. 56
 - ☒ 3. 59
 - ☒ 4. 58



Question ID : 630680504957
 Option 1 ID : 6306801973358
 Option 2 ID : 6306801973357
 Option 3 ID : 6306801973360
 Option 4 ID : 6306801973359

Status : Answered

Chosen Option : 1



Q.33 In the following question, select the related letter pair from the given alternatives. OPLN : MMHI :: ?

- Ans
- ☒ 1. FULM : ERWJ
 - ☒ 2. TEHV : RBDQ
 - ☒ 3. NOCP : JKUT
 - ☒ 4. AFIJ : YCED

Question ID : 630680504974
 Option 1 ID : 6306801973425
 Option 2 ID : 6306801973428
 Option 3 ID : 6306801973427
 Option 4 ID : 6306801973426
 Status : Answered
 Chosen Option : 2

Q.34 6 players R1, R2, R3, R4, R5 and R6 run on 6 different days of a week starting from Monday to Saturday. No two players run on the same day. Only one player runs between R5 and R6. No player runs after R3. Only one player runs between R3 and R6. R5 runs before R6 and after R1. R4 runs before R6. Who runs on Tuesday?

- Ans
- ☒ 1. R1
 - ☒ 2. R6
 - ☒ 3. R5
 - ☒ 4. R4

Question ID : 630680504959
 Option 1 ID : 6306801973365
 Option 2 ID : 6306801973366
 Option 3 ID : 6306801973367
 Option 4 ID : 6306801973368
 Status : Not Answered
 Chosen Option : -

Section : Domain Knowledge

Q.1 When designing a part for manufacturing using Fused Deposition Modeling (FDM), what aspect is crucial to enhance the part's strength and durability?

- Ans
- ☒ 1. Increasing the infill density of the part.
 - ☒ 2. Using a larger nozzle diameter.
 - ☒ 3. Selecting a faster printing speed.
 - ☒ 4. Choosing a higher nozzle temperature.

Question ID : 630680505460
 Option 1 ID : 6306801975372
 Option 2 ID : 6306801975370
 Option 3 ID : 6306801975369
 Option 4 ID : 6306801975371
 Status : Not Answered
 Chosen Option : -

Q.2 Which instrument is most appropriate for measuring the surface roughness of a finely machined component?

Ans ☒ 1. Optical profilometer

☐ 2. Micrometer

☐ 3. Dial indicator

☐ 4. Vernier caliper

Question ID : 630680505459

Option 1 ID : 6306801975367

Option 2 ID : 6306801975366

Option 3 ID : 6306801975365

Option 4 ID : 6306801975368

Status : Answered

Chosen Option : 1

Q.3 Which modern construction material is known for its high strength-to-weight ratio and is commonly used in innovative architectural designs?

Ans ☐ 1. Brick

☐ 2. Plywood

☒ 3. Carbon Fiber

☐ 4. Reinforced Concrete

Question ID : 630680505478

Option 1 ID : 6306801975441

Option 2 ID : 6306801975443

Option 3 ID : 6306801975442

Option 4 ID : 6306801975444

Status : Not Answered

Chosen Option : -

Q.4 Which government initiative in India primarily focuses on improving the technological infrastructure in education?

Ans ☒ 1. NDEAR

☐ 2. MDMS

☐ 3. SSA

☐ 4. RMSA

Question ID : 630680505471

Option 1 ID : 6306801975415

Option 2 ID : 6306801975414

Option 3 ID : 6306801975416

Option 4 ID : 6306801975413

Status : Not Answered

Chosen Option : -



Q.5 In a screw jack, if the pitch of the screw is increased while keeping the length of the handle constant, what effect does it have on the mechanical advantage of the jack?

- Ans**
- ☒ 1. No change in mechanical advantage
 - ☒ 2. Increases the mechanical advantage
 - ☒ 3. Decreases the mechanical advantage
 - ☒ 4. The mechanical advantage becomes unpredictable

Question ID : 630680505406
 Option 1 ID : 6306801975156
 Option 2 ID : 6306801975153
 Option 3 ID : 6306801975154
 Option 4 ID : 6306801975155

Status : Not Answered

Chosen Option : -

Q.6 During the heat treatment of a medium-carbon steel, which of the following microstructural changes is most likely to occur when the steel is rapidly cooled (quenched) from the austenitizing temperature?

- Ans**
- ☒ 1. The formation of pearlite and ferrite
 - ☒ 2. The retention of austenite
 - ☒ 3. The formation of martensite
 - ☒ 4. The formation of cementite and ferrite

Question ID : 630680505445
 Option 1 ID : 6306801975312
 Option 2 ID : 6306801975310
 Option 3 ID : 6306801975309
 Option 4 ID : 6306801975311

Status : Answered

Chosen Option : 3

Q.7 Consider a steady-state, steady-flow process in a control volume. If the mass flow rate entering the control volume is 5 kg/s, what is the mass flow rate exiting the control volume?

- Ans**
- ☒ 1. 0 kg/s
 - ☒ 2. 5 kg/s
 - ☒ 3. 10 kg/s
 - ☒ 4. -10 kg/s



Question ID : 630680505427
 Option 1 ID : 6306801975238
 Option 2 ID : 6306801975239
 Option 3 ID : 6306801975240
 Option 4 ID : 6306801975237

Status : Answered

Chosen Option : 2



Q.8 In a vibration isolation system using damping, which of the following statements best describes the effect of increasing the damping ratio beyond the critical damping ratio?

- Ans**
- ☒ 1. It has no significant effect on the isolation performance.
 - ☒ 2. It causes the system to become underdamped, leading to increased oscillations.
 - ☒ 3. It maximizes energy dissipation and improves isolation efficiency.
 - ☒ 4. It reduces the speed of response of the system, leading to poorer isolation.

Question ID : 630680505417
 Option 1 ID : 6306801975198
 Option 2 ID : 6306801975200
 Option 3 ID : 6306801975199
 Option 4 ID : 6306801975197
 Status : Not Answered
 Chosen Option : -

Q.9 Which of the following is a practical application of the principle of induced EMF?

- Ans**
- ☒ 1. Illuminating a room with LED lighting
 - ☒ 2. Recharging a smartphone using a wireless charger
 - ☒ 3. Cooling a room using an air conditioner
 - ☒ 4. Heating food in a microwave oven

Question ID : 630680505473
 Option 1 ID : 6306801975423
 Option 2 ID : 6306801975421
 Option 3 ID : 6306801975422
 Option 4 ID : 6306801975424
 Status : Not Answered
 Chosen Option : -

Q.10 What distinct feature of Kaplan turbines contributes to their high efficiency in low head, high flow hydroelectric power plants?

- Ans**
- ☒ 1. Adjustable wicket gates and fixed runner blades.
 - ☒ 2. Fixed blades and a simple runner design.
 - ☒ 3. Adjustable runner blades and fixed wicket gates.
 - ☒ 4. Both adjustable runner blades and wicket gates.

Question ID : 630680505443
 Option 1 ID : 6306801975301
 Option 2 ID : 6306801975304
 Option 3 ID : 6306801975302
 Option 4 ID : 6306801975303
 Status : Not Answered
 Chosen Option : -

Q.11 What is the significance of 'Scoping' in the EIA process?

Ans ☒ 1. It involves the actual mitigation of environmental impacts

☒ 2.

It identifies the key issues and impacts to be focused on in the EIA

☒ 3. It determines the project's financial viability

☒ 4.

It is the final stage where the decision is made whether to approve the project or not

Question ID : 630680505469

Option 1 ID : 6306801975407

Option 2 ID : 6306801975405

Option 3 ID : 6306801975406

Option 4 ID : 6306801975408

Status : Answered

Chosen Option : 2

Q.12 In construction planning, what is the primary purpose of a Gantt Chart?

Ans ☒ 1.

To visualize the project schedule, showing the start and finish dates of elements.

☒ 2. To allocate budget for different phases of construction.

☒ 3. To assess environmental impact of the construction project.

☒ 4. To provide a detailed architectural design of the project.

Question ID : 630680505481

Option 1 ID : 6306801975453

Option 2 ID : 6306801975454

Option 3 ID : 6306801975455

Option 4 ID : 6306801975456

Status : Answered

Chosen Option : 1

Q.13 What principle is primarily used in wind turbines to convert wind energy into electrical energy?

Ans ☒ 1. Piezoelectric effect

☒ 2. Chemical reaction

☒ 3. Electromagnetic induction

☒ 4. Photovoltaic effect

Question ID : 630680505485

Option 1 ID : 6306801975472

Option 2 ID : 6306801975470

Option 3 ID : 6306801975469

Option 4 ID : 6306801975471

Status : Answered

Chosen Option : 4



Q.14 A composite wall consists of three different materials with thermal conductivities k_1 , k_2 and k_3 & thickness d_1 , d_2 and d_3 respectively. If the wall is subject to steady heat conduction, which of the following expressions represents the overall heat transfer coefficient (U) of the wall?

Ans

✗ 1. $U = \frac{k_1 + k_2 + k_3}{d_1 + d_2 + d_3}$

✓ 2. $U = \frac{1}{\frac{d_1}{k_1} + \frac{d_2}{k_2} + \frac{d_3}{k_3}}$

✗ 3. $U = \frac{1}{d_1 + d_2 + d_3} \times (k_1 + k_2 + k_3)$

✗ 4. $U = \frac{k_1 \times k_2 \times k_3}{d_1 + d_2 + d_3}$

Question ID : 630680505438
Option 1 ID : 6306801975283
Option 2 ID : 6306801975282
Option 3 ID : 6306801975281
Option 4 ID : 6306801975284

Status : Answered

Chosen Option : 2

Q.15 Consider a thin-walled open section beam shaped like a 'U'. If a vertical shear force is applied at the centroid of its cross-section, how will the beam react?

Ans

✗ 1. It will bend without twisting

✓ 2. It will bend and twist

✗ 3. It will neither bend nor twist

✗ 4. It will twist without bending

Question ID : 630680505408
Option 1 ID : 6306801975163
Option 2 ID : 6306801975164
Option 3 ID : 6306801975161
Option 4 ID : 6306801975162

Status : Not Answered

Chosen Option : -

Q.16 At a given pressure, the compressibility factor (Z) of a real gas increases with an increase in temperature. This observation suggests that:

Ans

✓ 1. the gas behaves more ideally at higher temperatures.

✗ 2.

temperature has no significant impact on the real gas behavior.

✗ 3. the gas will liquefy at higher temperatures.

✗ 4. the gas behaves less ideally at higher temperatures.

Question ID : 630680505437
Option 1 ID : 6306801975277
Option 2 ID : 6306801975280
Option 3 ID : 6306801975279
Option 4 ID : 6306801975278

Status : Not Answered

Chosen Option : -



Q.17 A ball of mass 0.5 kg moving at 10 m/s collides with another stationary ball of mass 1 kg and comes to a complete stop. What is the velocity of the second ball post-collision? (Assume an elastic collision)

- Ans**
- ✓ 1. 5 m/s
 - ✗ 2. 10 m/s
 - ✗ 3. 7.5 m/s
 - ✗ 4. 15 m/s

Question ID : 630680505428
 Option 1 ID : 6306801975244
 Option 2 ID : 6306801975242
 Option 3 ID : 6306801975243
 Option 4 ID : 6306801975241
 Status : Not Answered
 Chosen Option : -

Q.18 A belt-driven system consists of two pulleys, one with a radius of 0.5 meters and the other with a radius of 0.2 meters. If the larger pulley rotates at 60 RPM (revolutions per minute), and the tension in the belt remains constant, what is the rotational speed of the smaller pulley?

- Ans**
- ✗ 1. 60 RPM
 - ✓ 2. 150 RPM
 - ✗ 3. 120 RPM
 - ✗ 4. 24 RPM

Question ID : 630680505404
 Option 1 ID : 6306801975146
 Option 2 ID : 6306801975148
 Option 3 ID : 6306801975147
 Option 4 ID : 6306801975145
 Status : Answered
 Chosen Option : 2

Q.19 A fluid with a density of 1000 kg/m^3 flows steadily through a horizontal pipe with a change in diameter. At a certain section, the diameter is 0.1 m, and the pressure is 200 kPa. If the velocity at this section is 2 m/s, what is the pressure at another section where the diameter is 0.2 m, assuming inviscid flow and using the Bernoulli equation? (May choose the nearest value)

- Ans**
- ✗ 1. 198.7 kPa
 - ✗ 2. 196.5 kPa
 - ✗ 3. 199.0 kPa
 - ✓ 4. 201.8 kPa

Question ID : 630680505424
 Option 1 ID : 6306801975226
 Option 2 ID : 6306801975225
 Option 3 ID : 6306801975227
 Option 4 ID : 6306801975228
 Status : Answered
 Chosen Option : 4



Q.20 When designing a gating system for a sand-casting process, to minimize turbulence and facilitate smooth flow of the molten metal, the gating system should be designed such that:

Ans ☒ 1.

The cross-sectional area of the runners is larger than the sprue and smaller than the gates.

☒ 2.

The cross-sectional area of the sprue, runners, and gates are all equal.

☒ 3.

The cross-sectional area of the sprue is larger than that of the runners and gates.

☒ 4.

The cross-sectional area of the sprue is smaller than that of the runners and gates.

Question ID : 630680505447

Option 1 ID : 6306801975317

Option 2 ID : 6306801975318

Option 3 ID : 6306801975320

Option 4 ID : 6306801975319

Status : Not Answered

Chosen Option : -

Q.21 In the context of Indian manufacturing industries, which standard is adopted for implementing an effective environmental management system (EMS), analogous to the ISO 14001?

Ans ☒ 1. ISO 14001:2017

☒ 2. ISO 14001:2015

☒ 3. ISO 9001:2015

☒ 4. ISO 15001:2016

Question ID : 630680505467

Option 1 ID : 6306801975400

Option 2 ID : 6306801975399

Option 3 ID : 6306801975397

Option 4 ID : 6306801975398

Status : Not Answered

Chosen Option : -

Q.22 In a typical hydrogen-oxygen fuel cell, what occurs at the cathode?

Ans ☒ 1. Protons (H^+) are converted into hydrogen gas (H_2).

☒ 2.

Water (H_2O) is electrolyzed to produce oxygen and hydrogen ions.

☒ 3. Hydrogen gas (H_2) is oxidized to produce electrons.

☒ 4.

Oxygen gas (O_2) reacts with electrons and hydrogen ions to form water.

Question ID : 630680505486

Option 1 ID : 6306801975473

Option 2 ID : 6306801975475

Option 3 ID : 6306801975476

Option 4 ID : 6306801975474

Status : Not Answered

Chosen Option : -



Q.23 In an air-standard Otto cycle, how does the specific heat ratio (γ) of the working fluid affect the thermal efficiency of the cycle?

- Ans
- ☒ 1. The efficiency is independent of γ .
 - ☒ 2. An increase in γ increases the efficiency.
 - ☒ 3. An increase in γ decreases the efficiency.
 - ☒ 4.

The efficiency initially increases with γ but then decreases after a certain point.

Question ID : 630680505441
 Option 1 ID : 6306801975295
 Option 2 ID : 6306801975293
 Option 3 ID : 6306801975294
 Option 4 ID : 6306801975296
 Status : Answered
 Chosen Option : 2

Q.24 What impact does effective site management have on construction productivity?

- Ans
- ☒ 1. Increases productivity by optimizing resource allocation
 - ☒ 2.
 - No impact, as productivity is solely dependent on worker skills
 - ☒ 3. Reduces productivity by limiting worker autonomy
 - ☒ 4. Decreases productivity due to increased overhead costs

Question ID : 630680505483
 Option 1 ID : 6306801975463
 Option 2 ID : 6306801975464
 Option 3 ID : 6306801975461
 Option 4 ID : 6306801975462
 Status : Not Answered
 Chosen Option : -

Q.25 In a vacuum, two parallel wires carrying currents I_1 and I_2 are separated by a distance d . If the currents are flowing in opposite directions, what is the force per unit length between the wires?

- Ans
- ☒ 1. $\frac{\mu_0 I_1 I_2}{\pi d}$ directed towards each other
 - ☒ 2. $\frac{\mu_0 I_1 I_2}{2\pi d}$ directed away from each other
 - ☒ 3. $\frac{\mu_0 I_1 I_2}{4\pi d}$ directed away from each other
 - ☒ 4. $\frac{\mu_0 I_1 I_2}{2\pi d}$ directed towards each other

Question ID : 630680505472
 Option 1 ID : 6306801975417
 Option 2 ID : 6306801975420
 Option 3 ID : 6306801975418
 Option 4 ID : 6306801975419
 Status : Not Answered
 Chosen Option : -



Q.26 Which statement best explains why real gases deviate from the ideal gas law at high pressures?

Ans ☒ 1. Intermolecular forces become negligible at high pressures.

☒ 2.

The kinetic energy of particles is proportional to temperature.

☒ 3. Gas particles occupy a negligible volume.

☒ 4.

Gas particles have finite volume and experience intermolecular attractions.

Question ID : 630680505434

Option 1 ID : 6306801975267

Option 2 ID : 6306801975265

Option 3 ID : 6306801975268

Option 4 ID : 6306801975266

Status : Not Answered

Chosen Option : -

Q.27 Which of the following is a limitation of the traditional assignment model in operations research?

Ans ☒ 1. It can only be applied to maximize the objective function.

☒ 2.

It cannot handle the assignment of tasks to more agents than tasks.

☒ 3. It requires the use of non-linear programming for solution.

☒ 4. It can only handle problems with non-integer solutions.

Question ID : 630680505465

Option 1 ID : 6306801975391

Option 2 ID : 6306801975389

Option 3 ID : 6306801975390

Option 4 ID : 6306801975392

Status : Not Answered

Chosen Option : -

Q.28 In a three-element rectangular strain gauge rosette, one gauge is aligned along the x-axis (ϵ_x), the second along the y-axis (ϵ_y), and the third at a 45-degree angle (ϵ_{45}). If the measured strains are $\epsilon_x = 300$ microstrain, $\epsilon_y = 100$ microstrain, and no shear strain is present, what is the strain measured by the 45-degree gauge?

Ans ☒ 1. 250 microstrain

☒ 2. 300 microstrain

☒ 3. 150 microstrain

☒ 4. 200 microstrain

Question ID : 630680505410

Option 1 ID : 6306801975171

Option 2 ID : 6306801975172

Option 3 ID : 6306801975169

Option 4 ID : 6306801975170

Status : Answered

Chosen Option : 3



Q.29 In metal forming processes, increasing the working temperature of a metal generally leads to:

Ans

✓ 1. Decreased yield strength and increased ductility.

✗ 2. Decreased yield strength and decreased ductility.

✗ 3.

No significant change in yield strength but increased brittleness.

✗ 4. Increased yield strength and decreased ductility.

Question ID : 630680505448

Option 1 ID : 6306801975324

Option 2 ID : 6306801975322

Option 3 ID : 6306801975321

Option 4 ID : 6306801975323

Status : Answered

Chosen Option : 1

Q.30 Which material is commonly used in high-temperature solar thermal energy storage systems for its ability to store a large amount of heat?

Ans

✗ 1. Silicon

✓ 2. Molten salts

✗ 3. Lithium-ion batteries

✗ 4. Water

Question ID : 630680505484

Option 1 ID : 6306801975466

Option 2 ID : 6306801975468

Option 3 ID : 6306801975465

Option 4 ID : 6306801975467

Status : Answered

Chosen Option : 1

Q.31 Which of the following pollutants is primarily responsible for the depletion of the ozone layer in the Earth's atmosphere?

Ans

✓ 1. Chlorofluorocarbons

✗ 2. Carbon dioxide

✗ 3. Methane

✗ 4. Nitrous oxide

Question ID : 630680505468

Option 1 ID : 6306801975404

Option 2 ID : 6306801975403

Option 3 ID : 6306801975402

Option 4 ID : 6306801975401

Status : Answered

Chosen Option : 1



Q.32 If a cylindrical shaft is designed with a basic size of 50 mm and a tolerance of ± 0.02 mm, and the hole is designed with a basic size of 50.02 mm and a tolerance of ± 0.02 mm, the type of fit most likely to be achieved is:

- Ans
- ☒ 1. Clearance fit
 - ☒ 2. Interference fit
 - ☒ 3. Loose fit
 - ☒ 4. Transition fit

Question ID : 630680505456
 Option 1 ID : 6306801975355
 Option 2 ID : 6306801975353
 Option 3 ID : 6306801975354
 Option 4 ID : 6306801975356
 Status : Answered
 Chosen Option : 1

Q.33 Why is preheating necessary before welding high-carbon steels?

- Ans
- ☒ 1. To decrease the cooling rate after welding.
 - ☒ 2. To increase the speed of welding.
 - ☒ 3. To enhance the magnetic properties of the steel.
 - ☒ 4. To reduce the electrical resistance of the steel.

Question ID : 630680505450
 Option 1 ID : 6306801975330
 Option 2 ID : 6306801975332
 Option 3 ID : 6306801975329
 Option 4 ID : 6306801975331
 Status : Answered
 Chosen Option : 1

Q.34 In a six-cylinder engine with all cylinders in a single plane, the primary balancing of reciprocating masses is perfect. If the mass of one reciprocating part is 0.5 kg and its stroke is 100 mm, what is the total primary unbalanced force at 4000 RPM? (May choose the nearest value)

- Ans
- ☒ 1. 0
 - ☒ 2. 628.32 N
 - ☒ 3. 1
 - ☒ 4. 314.16 N

Question ID : 630680505415
 Option 1 ID : 6306801975189
 Option 2 ID : 6306801975192
 Option 3 ID : 6306801975190
 Option 4 ID : 6306801975191
 Status : Answered
 Chosen Option : 1

Q.35 _____ is a process where small pieces from the tool rake face material are adhered to the flowing chip and thus removed from the surface.

- Ans
- ☒ 1. Adhesive wear
 - ☒ 2. Flank wear
 - ☒ 3. Crater wear
 - ☒ 4. Diffusion wear

Question ID : 630680505453
 Option 1 ID : 6306801975344
 Option 2 ID : 6306801975342
 Option 3 ID : 6306801975341
 Option 4 ID : 6306801975343
 Status : Not Answered
 Chosen Option : -

Q.36 In a bolted joint subjected to fluctuating tensile loads, which of the following factors does not significantly influence the fatigue life of the bolt?

- Ans
- ☒ 1. Surface finish of the bolt
 - ☒ 2. Bolt material's ultimate tensile strength
 - ☒ 3. Size of the washer used with the bolt
 - ☒ 4. Magnitude of preload applied to the bolt

Question ID : 630680505423
 Option 1 ID : 6306801975221
 Option 2 ID : 6306801975224
 Option 3 ID : 6306801975223
 Option 4 ID : 6306801975222
 Status : Answered
 Chosen Option : 4

Q.37 A damped harmonic oscillator with a damping ratio of 0.5 is subjected to a sinusoidal force of frequency equal to its natural frequency. What will be the phase difference between the applied force and the displacement response of the system?

- Ans
- ☒ 1. 180 degrees
 - ☒ 2. 45 degrees
 - ☒ 3. 30 degrees
 - ☒ 4. 0 degrees

Question ID : 630680505419
 Option 1 ID : 6306801975208
 Option 2 ID : 6306801975207
 Option 3 ID : 6306801975206
 Option 4 ID : 6306801975205
 Status : Not Answered
 Chosen Option : -



Q.38 In a four-bar linkage ABCD, where AB is the fixed link, BC is the coupler, and CD is the output link, if the input link AB rotates at a constant angular velocity of $\omega_{AB} = 10 \text{ rad/s}$, and at the instant of interest the angle $\angle ABC$ is 60° with $|AB|=|BC|=|CD|=1 \text{ m}$, what is the angular velocity of the output link CD (ω_{CD})?

- Ans
- ☒ 1. 10 rad/s
 - ☐ 2. 15 rad/s
 - ☐ 3. 5 rad/s
 - ☐ 4. 20 rad/s

Question ID : 630680505413
 Option 1 ID : 6306801975181
 Option 2 ID : 6306801975182
 Option 3 ID : 6306801975184
 Option 4 ID : 6306801975183

Status : Not Answered

Chosen Option : -

Q.39 Consider a double pendulum consisting of two masses m_1 and m_2 connected by massless rods of lengths l_1 and l_2 respectively. Using Lagrange's equation, which of the following represents a part of the Lagrangian for this system?

- Ans
- ☐ 1. $L = m_1 gl_1 \sin \theta_1 + m_2 gl_2 \sin \theta_2$
 - ☒ 2. $L = \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + m_2 gl_2 \cos \theta_2$
 - ☐ 3. $L = m_1 gl_1 \sin \theta_1 + m_2 gl_2 \cos \theta_2$
 - ☐ 4. $L = \frac{1}{2} m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 l_2^2 \dot{\theta}_2^2$

Question ID : 630680505405
 Option 1 ID : 6306801975152
 Option 2 ID : 6306801975149
 Option 3 ID : 6306801975151
 Option 4 ID : 6306801975150

Status : Not Answered

Chosen Option : -

Q.40 In object-oriented programming (OOP), which of the following concepts allows different classes to be treated as instances of the same superclass?

- Ans
- ☐ 1. Encapsulation
 - ☐ 2. Abstraction
 - ☒ 3. Polymorphism
 - ☐ 4. Inheritance

Question ID : 630680505476
 Option 1 ID : 6306801975435
 Option 2 ID : 6306801975434
 Option 3 ID : 6306801975436
 Option 4 ID : 6306801975433

Status : Not Answered

Chosen Option : -



Q.41 In a wind tunnel experiment, air flows over a 2 m long flat plate. If the Reynolds number at the trailing edge of the plate is 5×10^6 and the kinematic viscosity of air is $1.5 \times 10^{-5} \text{ m}^2/\text{s}$, what is the free stream velocity of the air? (May choose the nearest value)

- Ans**
- ☒ 1. 30 m/s
 - ☒ 2. 63.72 m/s
 - ☒ 3. 53.03 m/s
 - ☒ 4. 37.5 m/s

Question ID : 630680505426
 Option 1 ID : 6306801975233
 Option 2 ID : 6306801975236
 Option 3 ID : 6306801975235
 Option 4 ID : 6306801975234
 Status : Answered
 Chosen Option : 4

Q.42 Which of the following mechanisms primarily contributes to particle bonding during the sintering process in powder metallurgy?

- Ans**
- ☒ 1. Chemical reactions between different powders
 - ☒ 2. Diffusion of atoms across particle boundaries
 - ☒ 3. Mechanical interlocking of particles
 - ☒ 4. Melting and solidification of the powder

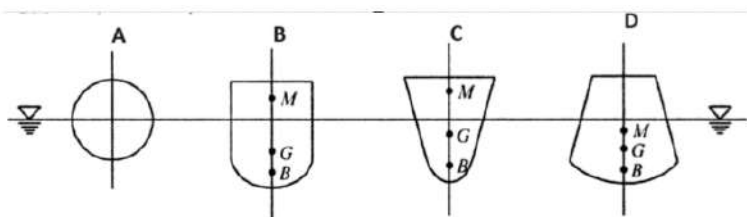
Question ID : 630680505449
 Option 1 ID : 6306801975326
 Option 2 ID : 6306801975325
 Option 3 ID : 6306801975328
 Option 4 ID : 6306801975327
 Status : Not Answered
 Chosen Option : -

Q.43 Which of the following file systems is specifically designed for flash storage devices like USB flash drives and SD cards?

- Ans**
- ☒ 1. ext4
 - ☒ 2. FAT32
 - ☒ 3. NTFS
 - ☒ 4. exFAT

Question ID : 630680505477
 Option 1 ID : 6306801975440
 Option 2 ID : 6306801975438
 Option 3 ID : 6306801975439
 Option 4 ID : 6306801975437
 Status : Not Answered
 Chosen Option : -

Q.44



Choose the correct pair for the given figure depicting the stability of different forms of floating vessel?

Ans

✗ 1.

A=Zero stability, B=poor stability, C=Better stability, D=Negative stability

✗ 2.

A=Zero stability, B=Better stability, C=Good stability, D=Doubtful stability

✗ 3.

A=Doubtful stability, B=Good stability, C=Better stability, D=Zero stability

✓ 4.

A=Zero stability, B=Good stability, C=Better stability, D=Doubtful stability

Question ID : 630680505429

Option 1 ID : 6306801975248

Option 2 ID : 6306801975247

Option 3 ID : 6306801975246

Option 4 ID : 6306801975245

Status : Answered

Chosen Option : 2

Q.45

A black surface and a grey surface are exposed to the same radiative environment. Which of the following statements is true regarding their emissivity (ϵ)?

Ans

✗ 1. $\epsilon_{black} = \epsilon_{grey}$

✗ 2. The relation between both surfaces cannot be expressed

✗ 3. $\epsilon_{black} < \epsilon_{grey}$

✓ 4. $\epsilon_{black} > \epsilon_{grey}$

Question ID : 630680505433

Option 1 ID : 6306801975262

Option 2 ID : 6306801975264

Option 3 ID : 6306801975263

Option 4 ID : 6306801975261

Status : Answered

Chosen Option : 4

Q.46

If a sphere with a thermal diffusivity of $1 \times 10^{-6} \text{ m}^2/\text{s}$ reaches a Fourier number of 0.15 in 500 seconds, what is its characteristic dimension (radius)?

Ans

✗ 1. 0.2 m

✗ 2. 5.8 m

✗ 3. 0.15 m

✓ 4. 0.05 m

Question ID : 630680505432

Option 1 ID : 6306801975259

Option 2 ID : 6306801975257

Option 3 ID : 6306801975258

Option 4 ID : 6306801975260

Status : Answered

Chosen Option : 4



Q.47 Which of the following is true statement for Lagrange's (Indirect Approach)?

- Ans
- ☒ 1. All forces considered.
 - ☒ 2. All forces handled via same expression.
 - ☒ 3. Free-body Diagrams useful.
 - ☒ 4. Intermediate forces less readily available.

Question ID : 630680505402
 Option 1 ID : 6306801975140
 Option 2 ID : 6306801975137
 Option 3 ID : 6306801975139
 Option 4 ID : 6306801975138
 Status : Not Answered
 Chosen Option : -

Q.48 A gear train consists of two spur gears. If the input gear has 24 teeth and rotates at 1200 RPM, and the output gear has 72 teeth, what is the output speed in RPM?

- Ans
- ☒ 1. 3600 rpm
 - ☒ 2. 2400 rpm
 - ☒ 3. 400 rpm
 - ☒ 4. 800 rpm

Question ID : 630680505414
 Option 1 ID : 6306801975185
 Option 2 ID : 6306801975188
 Option 3 ID : 6306801975186
 Option 4 ID : 6306801975187
 Status : Answered
 Chosen Option : 3

Q.49 In building construction, what is the primary purpose of using a damp-proof course (DPC)?

- Ans
- ☒ 1. To provide insulation to the building
 - ☒ 2. To prevent water seepage into the building from the ground
 - ☒ 3. To enhance the aesthetic appeal of the building
 - ☒ 4. To support the structural integrity of the building

Question ID : 630680505479
 Option 1 ID : 6306801975446
 Option 2 ID : 6306801975447
 Option 3 ID : 6306801975448
 Option 4 ID : 6306801975445
 Status : Answered
 Chosen Option : 4

Q.50 Which of the following is considered an output peripheral in a computer system?

- Ans
- ☐ 1. Mouse
 - ☒ 2. Monitor
 - ☐ 3. Hard Drive
 - ☐ 4. Keyboard

Question ID : 630680505475
Option 1 ID : 6306801975430
Option 2 ID : 6306801975431
Option 3 ID : 6306801975429
Option 4 ID : 6306801975432
Status : Answered
Chosen Option : 2

Q.51 In the tendering process, what is the primary purpose of a Request for Proposal (RFP)?

- Ans
- ☐ 1. To announce the project budget to potential contractors.
 - ☐ 2. To request a detailed project plan from potential contractors.
 - ☒ 3. To invite potential contractors to submit proposals for a project.
 - ☐ 4. To finalize the contract terms with the chosen contractor.

Question ID : 630680505482
Option 1 ID : 6306801975460
Option 2 ID : 6306801975457
Option 3 ID : 6306801975458
Option 4 ID : 6306801975459
Status : Not Answered
Chosen Option : -

Q.52 A company uses an Economic Order Quantity (EOQ) model for inventory control. The annual demand for their product is 10,000 units, the ordering cost per order is Rs.100, and the holding cost per unit per year is Rs.2. What is the EOQ for this product? (May select the most closest value)

- Ans
- ☐ 1. 100 units
 - ☐ 2. 2000 units
 - ☐ 3. 50 units
 - ☒ 4. 1000 units

Question ID : 630680505463
Option 1 ID : 6306801975382
Option 2 ID : 6306801975384
Option 3 ID : 6306801975381
Option 4 ID : 6306801975383
Status : Answered
Chosen Option : 4



Q.53 In a Rankine cycle, the implementation of regeneration through feedwater heating:

Ans ☒ 1.

Decreases the average temperature of heat addition, thereby reducing cycle efficiency.

☒ 2.

Has no impact on the average temperature of heat addition but increases the total heat input.

☒ 3.

Increases the total heat input, thereby decreasing the thermal efficiency of the cycle.

☒ 4.

Increases the average temperature of heat addition, thereby increasing cycle efficiency.

Question ID : 630680505440

Option 1 ID : 6306801975291

Option 2 ID : 6306801975289

Option 3 ID : 6306801975290

Option 4 ID : 6306801975292

Status : Answered

Chosen Option : 4

Q.54 A vernier caliper has a main scale with a least count of 1 mm and a vernier scale with 10 divisions. If the 5th division of the vernier scale aligns with a main scale reading after the 20 mm mark, what is the precise measurement?

Ans ☒ 1. 20.50 mm

☒ 2. 20.05 mm

☒ 3. 20.25 mm

☒ 4. 20.15 mm

Question ID : 630680505457

Option 1 ID : 6306801975357

Option 2 ID : 6306801975358

Option 3 ID : 6306801975360

Option 4 ID : 6306801975359

Status : Answered

Chosen Option : 1

Q.55 A cylindrical rod made of a material with Poisson's ratio $\nu = 0.3$ is stretched such that its length increases by 5%. If the diameter of the rod was initially 10 mm, what is the new diameter of the rod after stretching?

Ans ☒ 1. 9.85 mm

☒ 2. 9.55 mm

☒ 3. 10.00 mm

☒ 4. 10.15 mm

Question ID : 630680505407

Option 1 ID : 6306801975159

Option 2 ID : 6306801975158

Option 3 ID : 6306801975160

Option 4 ID : 6306801975157

Status : Answered

Chosen Option : 1



Q.56 In the context of cement concrete, what is the primary purpose of adding aggregates?

Ans ☒ 1. To decrease the setting time of concrete.

☒ 2.

To enhance the compressive strength and durability of concrete.

☒ 3. To improve the color and texture of the finished surface.

☒ 4. To increase the thermal expansion of concrete.

Question ID : 630680505480

Option 1 ID : 6306801975449

Option 2 ID : 6306801975452

Option 3 ID : 6306801975451

Option 4 ID : 6306801975450

Status : Not Answered

Chosen Option : -

Q.57 What is the primary goal of the 'BharatNet' project in the context of e-Governance in India?

Ans ☒ 1. To establish a nationwide broadband network

☒ 2. To provide digital literacy to rural citizens

☒ 3. To digitize government records

☒ 4. To promote online financial transactions such as UPI

Question ID : 630680505470

Option 1 ID : 6306801975410

Option 2 ID : 6306801975411

Option 3 ID : 6306801975412

Option 4 ID : 6306801975409

Status : Answered

Chosen Option : 1



Q.58 In the design of mechanical components for dynamic loading, which expression correctly represents Goodman line for assessing fatigue failure?

Where K_f = Fatigue stress concentration factor

S_e = modified endurance limit

S_{yt} = Yield strength

S_{ut} = Ultimate strength in tension

σ_m = mean stress

σ_a = alternating stress

Ans

✗ 1. $\frac{K_f n_s \sigma_a}{S_e} + \left(\frac{n_s \sigma_m}{S_{ut}} \right)^2 = 1$

✗ 2. $\frac{K_f \sigma_a}{S_e} + \frac{\sigma_m}{S_{yt}} = \frac{\sigma_m}{n_s}$

✗ 3. $\frac{K_f \sigma_a}{S_e} + \left(\frac{n_s \sigma_m}{S_{ut}} \right)^2 = \frac{1}{n_s}$

✓ 4. $\frac{K_f \sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{n_s}$

Question ID : 630680505421

Option 1 ID : 6306801975215

Option 2 ID : 6306801975214

Option 3 ID : 6306801975216

Option 4 ID : 6306801975213

Status : Answered

Chosen Option : 1

Q.59 On an S-N diagram, what does a flatter slope in the high-cycle fatigue region (beyond 10^3 cycles) typically indicate about a material's fatigue behavior?

Ans

✓ 1. Greater resistance to fatigue failure

✗ 2. Higher sensitivity to stress concentrations

✗ 3. Lower endurance limit

✗ 4.

Increased impact of environmental factors on fatigue strength

Question ID : 630680505422

Option 1 ID : 6306801975218

Option 2 ID : 6306801975217

Option 3 ID : 6306801975219

Option 4 ID : 6306801975220

Status : Not Answered

Chosen Option : -



Q.60 In a compression test conducted on a concrete specimen using a Universal Testing Machine, if the specimen fails at a load of 300 kN and the cross-sectional area of the specimen is 150 cm², what is the compressive strength of the concrete?

- Ans**
- ☒ 1. 200 kPa
 - ☒ 2. 20 MPa
 - ☒ 3. 2000 kPa
 - ☒ 4. 2 MPa

Question ID : 630680505411
 Option 1 ID : 6306801975173
 Option 2 ID : 6306801975175
 Option 3 ID : 6306801975176
 Option 4 ID : 6306801975174
 Status : Answered
 Chosen Option : 2

Q.61 When using steam tables to determine the properties of superheated steam at a given pressure and temperature, which of the following is a correct approach?

- Ans**
- ☒ 1. Use the saturated steam tables and adjust for superheat.
 - ☒ 2. Interpolate between two temperatures at constant pressure.
 - ☒ 3. Interpolate between two pressures at constant temperature.
 - ☒ 4. Directly read the values without any interpolation.

Question ID : 630680505436
 Option 1 ID : 6306801975274
 Option 2 ID : 6306801975276
 Option 3 ID : 6306801975275
 Option 4 ID : 6306801975273
 Status : Not Answered
 Chosen Option : -

Q.62 In a linear programming problem, the feasible region:

- Ans**
- ☒ 1. is the set of all solutions that satisfy the constraint equations but not necessarily the non-negativity restrictions.
 - ☒ 2. consists of only the boundary lines formed by the constraints.
 - ☒ 3. is defined by the intersection of all the constraint inequalities and the non-negativity restrictions.
 - ☒ 4. represents all possible values that satisfy the objective function.

Question ID : 630680505464
 Option 1 ID : 6306801975387
 Option 2 ID : 6306801975386
 Option 3 ID : 6306801975388
 Option 4 ID : 6306801975385
 Status : Answered
 Chosen Option : 3

Q.63 A typical stress-strain curve for a ductile material shows regions of elastic and plastic deformation. At a certain point on the curve, the material exhibits maximum stress before necking occurs. This point is known as:

- Ans**
- ☒ 1. Fracture point
 - ☒ 2. Proportional limit
 - ☒ 3. Yield point
 - ☒ 4. Ultimate tensile strength

Question ID : 630680505444
 Option 1 ID : 6306801975306
 Option 2 ID : 6306801975307
 Option 3 ID : 6306801975305
 Option 4 ID : 6306801975308

Status : Answered

Chosen Option : 4

Q.64 In aggregate production planning, which strategy is most effective for a manufacturing company facing highly variable demand while aiming to minimize changes in workforce levels?

- Ans**
- ☒ 1. Implementing a Just-In-Time (JIT) production system.
 - ☒ 2. Maintaining a constant production rate and using inventory to absorb fluctuations in demand.
 - ☒ 3. Providing the service or product at an earlier time period.
 - ☒ 4. Frequently adjusting production rates to match demand and hiring temporary staff as needed.

Question ID : 630680505461
 Option 1 ID : 6306801975376
 Option 2 ID : 6306801975373
 Option 3 ID : 6306801975375
 Option 4 ID : 6306801975374

Status : Not Answered

Chosen Option : -

Q.65 Implementing the 5S methodology in a manufacturing setting primarily helps in:

- Ans**
- ☒ 1. Optimizing the supply chain and logistics.
 - ☒ 2. Enhancing employee skills and capabilities.
 - ☒ 3. Reducing lead time for product development.
 - ☒ 4. Improving workplace organization and efficiency.

Question ID : 630680505462
 Option 1 ID : 6306801975379
 Option 2 ID : 6306801975380
 Option 3 ID : 6306801975377
 Option 4 ID : 6306801975378

Status : Not Answered

Chosen Option : -

Q.66 According to Euler's theory, for a column with both ends pinned (hinged), if the length of the column is doubled and all other factors remain constant, how does the critical buckling load change?

- Ans
- ☒ 1. It increases by a factor of two
 - ☒ 2. It decreases by a factor of four
 - ☒ 3. It decreases by a factor of sixteen
 - ☒ 4. It remains unchanged

Question ID : 630680505409
Option 1 ID : 6306801975166
Option 2 ID : 6306801975165
Option 3 ID : 6306801975167
Option 4 ID : 6306801975168

Status : Answered

Chosen Option : 2

Q.67 In a Michelson interferometer, if the wavelength of the light source is halved, how does this affect the resolution of the system?

- Ans
- ☒ 1. The resolution is quadrupled.
 - ☒ 2. The resolution remains unchanged.
 - ☒ 3. The resolution is doubled.
 - ☒ 4. The resolution is halved.

Question ID : 630680505458
Option 1 ID : 6306801975363
Option 2 ID : 6306801975364
Option 3 ID : 6306801975362
Option 4 ID : 6306801975361

Status : Not Answered

Chosen Option : -

Q.68 Ultrasonic Machining (USM) is particularly effective for machining which type of materials?

- Ans
- ☒ 1. Soft, ductile materials
 - ☒ 2. Thermoplastics and polymers
 - ☒ 3. Hard and brittle materials such as ceramics
 - ☒ 4. High-strength ferrous metals

Question ID : 630680505454
Option 1 ID : 6306801975347
Option 2 ID : 6306801975345
Option 3 ID : 6306801975346
Option 4 ID : 6306801975348

Status : Answered

Chosen Option : 3



Q.69 Consider a multi-stage air compressor operating under ideal conditions. If each stage of the compressor is designed to have the same pressure ratio, which of the following statements is true regarding the overall efficiency of the compressor?

Ans  1.

The efficiency first increases and then decreases with the number of stages.

 2. The number of stages has no impact on the efficiency.

 3. The efficiency decreases as the number of stages increases.

 4. The efficiency increases as the number of stages increases.

Question ID : 630680505439

Option 1 ID : 6306801975288

Option 2 ID : 6306801975285

Option 3 ID : 6306801975286

Option 4 ID : 6306801975287

Status : Not Answered

Chosen Option : -

Q.70 In a gyroscope, if the spin speed is doubled while keeping the mass and radius of the gyroscope wheel constant, what is the effect on the gyroscopic couple assuming no external forces are acting?

Ans  1. The gyroscopic couple is quadrupled

 2. The gyroscopic couple is halved

 3. The gyroscopic couple is doubled

 4. The gyroscopic couple remains the same

Question ID : 630680505412

Option 1 ID : 6306801975180

Option 2 ID : 6306801975177

Option 3 ID : 6306801975178

Option 4 ID : 6306801975179

Status : Answered

Chosen Option : 3

Q.71 Which material is most suitable for making a cutting tool used in high-speed machining of superalloys?

Ans  1. High-Speed Steel

 2. Polycrystalline Diamond

 3. Cubic Boron Nitride

 4. Cemented Carbide

Question ID : 630680505452

Option 1 ID : 6306801975339

Option 2 ID : 6306801975337

Option 3 ID : 6306801975340

Option 4 ID : 6306801975338

Status : Answered

Chosen Option : 4



Q.72 A cylindrical steel rod subjected to a tensile stress of 300 MPa has a yield strength in tension of 400 MPa and in compression of 380 MPa. According to the Maximum Shear Stress Theory (Tresca's Criterion), at what stress level will the rod begin to yield?

- Ans**
- ☒ 1. 400 MPa
 - ☒ 2. 300 MPa
 - ☒ 3. 380 MPa
 - ☒ 4. 190 MPa

Question ID : 630680505420
 Option 1 ID : 6306801975211
 Option 2 ID : 6306801975209
 Option 3 ID : 6306801975212
 Option 4 ID : 6306801975210

Status : Answered

Chosen Option : 1

Q.73 A model airplane is being tested in a wind tunnel to study its aerodynamic properties. If the airplane's speed (V), wing area (A), air density (ρ), and the coefficient of drag (C_d) are known, which of the following combinations would correctly yield the drag force (F_d) acting on the airplane using dimensional analysis?

- Ans**
- ☒ 1. $F_d = C_d \times 2\rho \times V^2 \times A$
 - ☒ 2. $F_d = \frac{1}{2} \times C_d \times \rho \times V^2 \times A$
 - ☒ 3. $F_d = \frac{1}{2} \times C_d \times \frac{\rho}{V} \times A$
 - ☒ 4. $F_d = C_d \times \frac{V}{\rho} \times A$

Question ID : 630680505425
 Option 1 ID : 6306801975231
 Option 2 ID : 6306801975229
 Option 3 ID : 6306801975230
 Option 4 ID : 6306801975232

Status : Answered

Chosen Option : 2

Q.74 In a one-dimensional steady-state heat conduction scenario, a rod of length L with thermal conductivity k experiences a temperature drop from T_1 to T_2 along its length. If the length of the rod is doubled and all other conditions remain unchanged, the new temperature gradient in the rod will be:

- Ans**
- ☒ 1. halved.
 - ☒ 2. quadrupled.
 - ☒ 3. unchanged.
 - ☒ 4. doubled.

Question ID : 630680505430
 Option 1 ID : 6306801975250
 Option 2 ID : 6306801975252
 Option 3 ID : 6306801975249
 Option 4 ID : 6306801975251

Status : Answered

Chosen Option : 3



Q.75 In surface grinding machines, which factor most significantly affects the surface finish of the workpiece?

- Ans
- ☒ 1. The hardness of the grinding wheel
 - ☐ 2. The grain size of the grinding wheel
 - ☐ 3. The speed of the grinding wheel
 - ☐ 4. The material of the workpiece

Question ID : 630680505451
Option 1 ID : 6306801975336
Option 2 ID : 6306801975335
Option 3 ID : 6306801975334
Option 4 ID : 6306801975333
Status : Answered
Chosen Option : 3

Q.76 A single degree of freedom system with a damping ratio of 0.05 is subjected to forced vibration at a frequency that is 1.5 times its natural frequency. If the system's natural frequency is 10 Hz, what is the frequency of the forced vibration?

- Ans
- ☐ 1. 10 Hz
 - ☐ 2. 25 Hz
 - ☐ 3. 20 Hz
 - ☒ 4. 15 Hz

Question ID : 630680505416
Option 1 ID : 6306801975195
Option 2 ID : 6306801975193
Option 3 ID : 6306801975194
Option 4 ID : 6306801975196
Status : Answered
Chosen Option : 4

Q.77 A long, straight fin with uniform cross-sectional area operates in a steady-state condition. If the thermal conductivity of the fin material increases, what is the impact on the fin's efficiency?

- Ans
- ☐ 1. The efficiency remains unchanged
 - ☐ 2. The efficiency decreases
 - ☐ 3. The efficiency first increases and then decreases
 - ☒ 4. The efficiency increases

Question ID : 630680505431
Option 1 ID : 6306801975254
Option 2 ID : 6306801975256
Option 3 ID : 6306801975253
Option 4 ID : 6306801975255
Status : Answered
Chosen Option : 3



Q.78 For an M/M/1 queuing system with an arrival rate of 6 customers per hour and a service rate of 10 customers per hour, what is the average number of customers in the system (L)?

- Ans
- ☒ 1. 2.4 customers
 - ☒ 2. 1.5 customers
 - ☒ 3. 3.0 customers
 - ☒ 4. 0.6 customers

Question ID : 630680505466
 Option 1 ID : 6306801975395
 Option 2 ID : 6306801975394
 Option 3 ID : 6306801975396
 Option 4 ID : 6306801975393
 Status : Answered
 Chosen Option : 2

Q.79 During an isothermal expansion of an ideal gas, the work done by the gas is given by:

- Ans
- ☒ 1. $W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$
 - ☒ 2. $W = \frac{1}{2} m(v_2^2 - v_1^2)$
 - ☒ 3. $W = nRT \ln \left(\frac{V_2}{V_1} \right)$
 - ☒ 4. $W = P(V_2 - V_1)$

Question ID : 630680505435
 Option 1 ID : 6306801975270
 Option 2 ID : 6306801975269
 Option 3 ID : 6306801975271
 Option 4 ID : 6306801975272
 Status : Answered
 Chosen Option : 3

Q.80 In a synchronous machine, when the load angle δ is zero, what is the condition of the machine?

- Ans
- ☒ 1. The machine is operating at unity power factor.
 - ☒ 2. The machine is in a state of instability.
 - ☒ 3. The machine is not delivering any real power.
 - ☒ 4. The machine is generating maximum power.

Question ID : 630680505474
 Option 1 ID : 6306801975427
 Option 2 ID : 6306801975425
 Option 3 ID : 6306801975426
 Option 4 ID : 6306801975428
 Status : Not Answered
 Chosen Option : -

Q.81 In the context of grinding wheels, how does the grain size of the abrasive material affect the surface finish and material removal rate (MRR) of the machining process?

Ans ☒ 1.

Coarser grain size results in a better surface finish and higher MRR.

☒ 2.

Finer grain size results in a better surface finish and higher MRR.

☒ 3.

Finer grain size results in a better surface finish but lower MRR.

☒ 4.

Coarser grain size results in a lower surface finish and lower MRR.

Question ID : 630680505455

Option 1 ID : 6306801975349

Option 2 ID : 6306801975352

Option 3 ID : 6306801975351

Option 4 ID : 6306801975350

Status : Answered

Chosen Option : 3

Q.82 A mechanical system exhibits critical damping. Which of the following best describes the time it takes for the system to return to equilibrium after being displaced?

Ans ☒ 1. Longer than an undamped system but with oscillations

☒ 2. Instantaneous

☒ 3. Shortest possible time without oscillating

☒ 4. Time is equal to that of an undamped system

Question ID : 630680505418

Option 1 ID : 6306801975204

Option 2 ID : 6306801975203

Option 3 ID : 6306801975202

Option 4 ID : 6306801975201

Status : Answered

Chosen Option : 3

Q.83 In a statically determinate planar frame, if one of the horizontal members is removed, how will it affect the structure's stability?

Ans ☒ 1. The change will not affect the structure's stability.

☒ 2. The structure will remain stable but more flexible.

☒ 3. The structure will become more rigid.

☒ 4. The structure will become unstable.

Question ID : 630680505403

Option 1 ID : 6306801975144

Option 2 ID : 6306801975142

Option 3 ID : 6306801975141

Option 4 ID : 6306801975143

Status : Not Answered

Chosen Option : -



Q.84 Two streams of moist air, Stream A and Stream B, are mixed. Stream A has a dry-bulb temperature of 30°C and a relative humidity of 60%, and Stream B has a dry-bulb temperature of 20°C and a relative humidity of 50%. Assuming no heat loss to the surroundings and a perfectly mixed final state, which of the following outcomes is most likely for the mixture?

Ans ☒ 1.

Dry-bulb temperature higher than 30°C, relative humidity lower than both streams

☒ 2.

Dry-bulb temperature between 20°C and 30°C, relative humidity higher than both streams

☒ 3.

Dry-bulb temperature between 20°C and 30°C, relative humidity higher than both streams

☒ 4.

Dry-bulb temperature between 20°C and 30°C, relative humidity lower than both streams

Question ID : 630680505442

Option 1 ID : 6306801975297

Option 2 ID : 6306801975299

Option 3 ID : 6306801975298

Option 4 ID : 6306801975300

Status : Answered

Chosen Option : 2

Q.85 Which of the following lost-core molding process is popularized by "The Ashland process" uses thermosetting hybrid system of phenolic resin and polyurethane as the binder?

Ans ☒ 1. Cold box

☒ 2. No-bake

☒ 3. Hot box

☒ 4. Shell molding

Question ID : 630680505446

Option 1 ID : 6306801975314

Option 2 ID : 6306801975316

Option 3 ID : 6306801975315

Option 4 ID : 6306801975313

Status : Answered

Chosen Option : 4



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